



**INTELLIGENT IOT-BASED MULTI-PARAMETER
CARDIO RESPIRATORY AND HEALTH MONITORING
WITH SMART HEALTH SCORING**



A PROJECT REPORT

Submitted by

KANISHKA L	2303811710622049
KEERTHIGA T	2303811710622052
LIBINA X	2303811710622058

*in partial fulfillment of the requirements for the award degree of
Bachelor of Engineering*

**ECC1375 - IOT BASED SYSTEM DESIGN
(PRBL)**

**DEPARTMENT OF ELECTRONICS AND
COMMUNICATION ENGINEERING**

**K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(AUTONOMOUS)**

SAMAYAPURAM – 621112

MAY 2026



**INTELLIGENT IOT-BASED MULTI-PARAMETER
CARDIO RESPIRATORY AND HEALTH MONITORING
WITH SMART HEALTH SCORING**



A PROJECT REPORT

Submitted by

KANISHKA L	2303811710622049
KEERTHIGA T	2303811710622052
LIBINA X	2303811710622058

*in partial fulfillment of the requirements for the award degree of
Bachelor of Engineering*

**ECC1375 - IOT BASED SYSTEM DESIGN
(PRBL)**

**DEPARTMENT OF ELECTRONICS AND
COMMUNICATION ENGINEERING**

**K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(AUTONOMOUS)**

SAMAYAPURAM – 621112

MAY 2026

**K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(AUTONOMOUS)**

SAMAYAPURAM - 621112

BONAFIDE CERTIFICATE

The work embodied in the present project report entitled “**INTELLIGENT IOT-BASED MULTI-PARAMETER CARDIO RESPIRATORY AND HEALTH MONITORING WITH SMART HEALTH SCORING**” is the bonafide work of Kanishka L , Keerthiga T , Libina X. The work reported herein is original, and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

Date of Viva Voce:

Mr. P.MANI, M.E.(Ph.D),

SUPERVISOR

Assistant Professor

Department of Electronics and

Communication Engineering

K. Ramakrishnan College of

Technology (Autonomous)

Samayapuram – 621112

Dr. S. SYEDAKBAR, M.E.,Ph.D.

HEAD OF THE DEPARTMENT

Assistant Professor

Department of Electronics and

Communication Engineering

K. Ramakrishnan College of

Technology (Autonomous)

Samayapuram – 621112

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

The rapid advancement of Internet of Things (IoT) technology has enabled efficient and continuous healthcare monitoring systems. Traditional monitoring of vital parameters such as heart rate and blood oxygen saturation (SpO₂) often requires frequent hospital visits and does not support real-time remote observation. This project proposes an IoT-Integrated Remote Health Surveillance System for non-invasive monitoring of heart rate and SpO₂ levels. The system uses sensors to collect physiological data and transmit it to a cloud platform for real-time monitoring and analysis. Healthcare professionals and caregivers can remotely access patient data through a web or mobile interface. The system also provides alert notifications when abnormal health conditions are detected. In addition to basic monitoring, the system includes stress detection for analyzing mental health conditions. It also estimates blood pressure and respiratory rate using sensor data analysis. A Smart Health Score is generated to provide an overall assessment of the user's health condition. This system improves accessibility, reduces hospital visits, and enables early detection of potential health risks.

Keywords:

IoT, Remote Health Monitoring, Heart Rate Monitoring, SpO₂ Measurement, Stress Detection, Blood Pressure Estimation, Respiratory Rate Estimation, Smart Health Score, Cloud-Based Healthcare, Non-Invasive Monitoring.

ACKNOWLEDGEMENT

We thank our **Dr. N. Vasudevan**, Principal, for his valuable suggestions and support during the course of my project work.

We thank our **Dr. S. Syedakbar**, Head of the Department, Department of Electronics and Communication Engineering, for providing his valuable suggestions and support during the course of my research work.

We wish to record my deep sense of gratitude and profound thanks to our Guide, **Mr. P. Mani**, Assistant Professor, Department of Electronics and Communication Engineering for his keen interest, inspiring guidance, constant encouragement with my work during all stages, to bring this thesis into fruition.

We are extremely indebted to our project coordinator **Mr. P. Mani**, Assistant Professor, Department of Electronics and Communication Engineering for his valuable suggestions and support during the course of my project work.

We also thank the faculty and non-teaching staff members of the Department of Electronics and Communication Engineering, K. Ramakrishnan College of Technology, Samayapuram, for their valuable support throughout the course of my project work.

Finally, we thank our parents, friends and our well wishes for their kind support.

SIGNATURE

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iii
	LIST OF FIGURES	viii
	LIST OF ABBREVIATIONS	ix
1	INTRODUCTION	1
	1.1 BACKGROUND	2
	1.2 OBJECTIVE OF THE PROJECT	3
	1.3 NOVELTY OF THE PROJECT	4
2	LITERATURE SURVEY	6
	2.1 SUMMARY OF LITERATURE	9
3	EXISTING AND PROPOSED METHOD	10
	3.1 EXISTING METHOD	10
	3.2 PROPOSED METHOD	13
	3.3 BLOCK DIAGRAM	17
	3.4 CIRCUIT DIAGRAM	20
	3.5 WORKING PRINCIPLE AND IMPLEMENTATION	23
	3.6 ADVANTAGES	26
	3.7 APPLICATIONS	28
4	HARDWARE AND SOFTWARE REQUIREMENTS	31
	4.1 HARDWARE REQUIREMENTS	31
	4.1.1 ESP32	31

4.1.2	PULSE OXIMETER SENSOR	32
4.1.3	OLED DISPLAY	33
4.1.4	POWER SUPPLY	34
4.2	SOFTWARE REQUIREMENTS	34
4.2.1	THINGSPEAK CLOUD PLATFORM	36
4.2.2	FIREBASE DATABASE	37
4.2.3	USER AUTHENTICATION INTERFACE	39
5	RESULTS AND DISCUSSION	42
5.1	HARDWARE PROTOTYPE	42
5.2	LIVE MONITORING SYSTEM	44
5.3	DISCUSSION	45
6	CONCLUSION	47
6.1	FUTURE ENHANCEMENTS	49
6.1.1	INTEGRATION OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING	49
6.1.2	ADDITION OF ADVANCED MEDICAL SENSORS	50
6.1.3	DEVELOPMENT OF A DEDICATED MOBILE APPLICATION	50
6.1.4	ENHANCED DATA SECURITY AND PRIVACY MECHANISMS	50
6.1.5	SCALABILITY AND MULTI- PATIENT MONITORING	51
6.1.6	ENERGY EFFICIENCY AND POWER OPTIMIZATION	51
6.1.7	INTEGRATION WITH GPS AND EMERGENCY RESPONSE SYSTEMS	51

6.1.8	CLOUD COMPUTING AND BIG DATA ANALYTICS	51
6.1.9	VOICE ASSISTANT AND SMART INTERACTION	52
6.1.10	INTEGRATION WITH SMART HEALTHCARE ECOSYSTEM	52
6.1.11	WEARABLE AND COMPACT DEVICE DEVELOPMENT	52
6.1.12	REAL-TIME VIDEO CONSULTATION INTEGRATION	52
6.2	LIMITATIONS OF THE PROPOSED WORK	53
6.2.1	KEY LIMITATIONS	53
6.2.2	EXPLANATION OF LIMITATIONS	53
6.3	SUMMARY OF THE PROJECT	55
7	REFERENCES	59

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
3.1	Existing Wearable Health Monitoring and Safety System	11
3.2	Proposed Method of Iot-Based Multi-Parameter Cardio Respiratory and Health Monitoring with Smart Health Scoring	15
3.3	Block Diagram of Proposed Smart Health Scoring System	18
3.4	Circuit Diagram of Proposed Smart Health Scoring System	21
4.1.1	ESP32 Microcontroller	31
4.1.2	Pulse Oximeter Sensor	32
4.1.3	EMG Sensor Module with Electrodes	33
4.2.1	ThinkSpeak Platform	36
4.2.2	Firestore Realtime Database	38
4.2.3	Live Monitoring Dashboard	40
5.1	Hardware Prototype of IoT-Based Wearable Safety System	42
5.2	Live Monitoring Dashboard Displaying Real-Time Physiological Parameters	44

LIST OF ABBREVIATIONS

BP	-	Blood Pressure
BPM	-	Beats Per Minute
ESP32	-	Espressif 32-Bit Microcontroller
GUI	-	Graphical User Interface
LED	-	Light Emitting Diode
OLED	-	Organic Light Emitting Diode
USB	-	Universal Serial Bus
Wi-Fi	-	Wireless Fidelity

CHAPTER 1

INTRODUCTION

Healthcare monitoring has become an important area of research due to the increasing number of patients suffering from chronic diseases and lifestyle-related health problems. Continuous monitoring of vital parameters such as heart rate, blood oxygen level, and respiration rate plays a crucial role in early diagnosis and prevention of serious medical conditions.

With the emergence of Internet of Things (IoT) technology, healthcare systems can now collect and transmit physiological data in real time using interconnected devices and sensors. IoT-based health monitoring systems allow doctors and caregivers to remotely observe patient health conditions without requiring constant hospital visits. These systems enhance patient safety, improve healthcare efficiency, and reduce medical costs.

In addition to EMG sensing, the system integrates other vital sensors such as heart rate and oxygen saturation monitors to enhance accuracy and reliability. The collected data is processed using a microcontroller, such as the ESP32, which enables efficient data handling, decision-making, and wireless communication. By combining multiple physiological parameters, the system ensures a more comprehensive understanding of the user's condition.

Modern healthcare systems are increasingly adopting smart technologies to improve patient monitoring and disease prevention. Wearable sensors and IoT-enabled devices allow continuous tracking of physiological parameters without disturbing the patient's daily routine. These technologies help in collecting accurate real-time health data, which can be analyzed to identify abnormal patterns and potential health risks. As a result, doctors can make faster and more informed medical decisions based on continuous patient data rather than occasional hospital measurements. Mobile applications provide users and doctors with instant notifications and health reports, enabling quick response during emergencies.

Remote health monitoring systems are particularly beneficial for elderly patients, individuals with chronic diseases, and people living in remote areas with limited access to healthcare facilities. By integrating cloud platforms and wireless communication technologies, patient health information can be securely stored and accessed anytime by authorized medical professionals. This ensures timely medical intervention in case of emergencies and reduces the burden on hospitals by enabling home-based healthcare monitoring.

Furthermore, integrating multiple health indicators such as stress level, blood pressure estimation, and respiratory rate provides a more comprehensive understanding of a person's overall health status. The introduction of a Smart Health Score helps in summarizing various physiological parameters into a single health index, making it easier for both patients and doctors to interpret health conditions quickly. Therefore, the proposed IoT-based health surveillance system aims to provide a reliable, cost-effective, and intelligent healthcare monitoring solution for modern medical needs.

1.1 BACKGROUND

Traditional healthcare monitoring methods rely on standalone medical devices such as pulse oximeters and heart rate monitors. These devices display readings locally and require patients to visit hospitals or diagnostic centers frequently for medical evaluation. Such methods are commonly used in clinical environments where medical professionals manually check and record patient data.

However, these conventional systems have several limitations. They do not support continuous monitoring of patient health conditions and are not suitable for long-term observation outside hospital environments. Patients must repeatedly visit healthcare facilities to measure their vital parameters, which can be inconvenient and time-consuming. Another major limitation of traditional systems is the lack of remote accessibility. The measured data is usually displayed only on the device and is not stored or transmitted to doctors or caregivers for further analysis. In emergency situations, this can increase the risk of serious health complications.

Recent developments in Internet of Things (IoT) technology and wearable sensors have enabled the creation of smart healthcare monitoring systems. These systems can continuously measure physiological parameters such as heart rate, oxygen saturation, and respiratory rate using compact sensors. The collected data can then be transmitted to cloud platforms through wireless networks for real-time monitoring.

With the integration of advanced analytics such as stress detection, blood pressure estimation, and respiratory monitoring, modern health surveillance systems provide a more comprehensive view of a patient's health condition. These smart systems allow doctors and caregivers to monitor patients remotely, generate alerts for abnormal readings, and ensure timely medical assistance, thereby improving the overall efficiency of healthcare services.

1.2 OBJECTIVE OF THE PROJECT

The main objective of this project is to develop an intelligent healthcare monitoring system using Internet of Things (IoT) technology that can continuously monitor important physiological parameters of a person. The system is designed to collect health data through sensors, process the information, and transmit it to a cloud platform for remote monitoring. By integrating multiple health monitoring features into a single system, the project aims to improve healthcare accessibility, support early disease detection, and reduce the need for frequent hospital visits.

The specific objectives of the project are:

- To design and develop a system capable of continuously monitoring heart rate and blood oxygen saturation (SpO₂) using non-invasive sensors.
- This allows users to track their vital health parameters in real time and detect abnormal conditions at an early stage.
- To integrate additional physiological sensors such as heart rate and oxygen saturation sensors to improve the accuracy and reliability of health monitoring.
- To transmit the collected physiological data to a cloud platform through IoT connectivity. This enables remote monitoring by doctors, caregivers, or family members through mobile devices or web dashboards.

- To enhance the system by estimating additional health indicators such as stress level, blood pressure, and respiratory rate using sensor data and signal analysis techniques. These parameters help provide a deeper understanding of the user's physical and mental health status.
- To develop a Smart Health Score that summarizes multiple health parameters into a single health index. This score helps users and healthcare professionals quickly evaluate the overall health condition of an individual.
- To design a compact, cost-effective, and easy-to-use monitoring device that can be used at home or in remote areas, making healthcare monitoring more accessible and convenient for users.

1.3 NOVELTY OF THE PROJECT

The novelty of the proposed project lies in integrating multiple health monitoring features into a single IoT-based platform.

Unlike conventional monitoring systems that only measure heart rate and SpO₂, the proposed system extends functionality by incorporating stress detection, blood pressure estimation, and respiratory rate monitoring. These parameters are analyzed together to generate a Smart Health Score that provides an overall assessment of the user's health condition.

The integration of IoT technology with cloud-based data storage also enhances the efficiency of healthcare monitoring. All the collected health data is stored securely in the cloud, allowing doctors and caregivers to access patient information from any location at any time. This feature supports remote healthcare services and helps in providing timely medical assistance when abnormal health conditions are detected.

Another unique aspect of the system is its ability to transmit real-time data to a cloud platform, enabling remote access for doctors and caregivers. The system also supports alert generation when abnormal health conditions are detected. This integration of multiple physiological parameters, remote monitoring, and health analytics makes the system more intelligent and efficient compared to existing healthcare monitoring solutions.

In addition, the system is designed to support continuous data logging and visualization. By analyzing historical health data, doctors can identify long-term trends and patterns in the patient's physiological parameters. This capability helps in improving medical diagnosis, monitoring treatment progress, and predicting potential health risks.

Another innovative feature of the proposed system is the generation of a Smart Health Score based on multiple physiological parameters. Instead of analyzing each health parameter separately, the system combines heart rate, SpO₂, stress level, blood pressure estimation, and respiratory rate into a single health indicator. This makes it easier for users and healthcare professionals to quickly understand the overall health status of the individual.

Another important novelty of the proposed system is the use of non-invasive sensing techniques for measuring vital health parameters. The system collects physiological signals from the user without causing discomfort or requiring complex medical procedures. This makes the device suitable for continuous daily monitoring and convenient for people of all age groups, especially elderly patients and individuals with chronic health conditions.

Another notable novelty of the proposed system is its capability to provide preventive healthcare support through continuous monitoring and intelligent data analysis. By constantly tracking multiple physiological parameters, the system can identify early signs of potential health issues before they become serious. This proactive approach helps users take necessary precautions and seek medical advice at the right time. As a result, the system not only supports real-time health monitoring but also contributes to improving overall health awareness and promoting better lifestyle management.

Finally, the proposed system focuses on developing a low-cost, compact, and portable healthcare monitoring device. By combining multiple monitoring functions into a single IoT-based platform, the system reduces the need for multiple medical devices. This approach not only lowers the overall cost but also makes the healthcare monitoring system more practical for home-based and remote healthcare applications.

CHAPTER 2

LITERATURE SURVEY

In 2026, R. Chauhan, P. Singh, A. Verma, and S. Tiwari in the paper “AI-Powered IoT Framework for Real-Time Respiratory Monitoring” proposed an intelligent framework that integrates Artificial Intelligence with IoT technology to monitor respiratory signals in real time. Their work emphasized advanced signal processing techniques and AI-based analytics for detecting abnormal breathing patterns. The study demonstrated improved accuracy in respiratory monitoring and highlighted the importance of intelligent healthcare systems for early disease detection and patient safety.

In 2025, A. Amir, M. Usman, H. Raza, and S. Khan in the paper “IoT-Based Health Monitoring Using Arduino and FPGA” developed a health monitoring system that integrates Arduino and FPGA platforms for efficient physiological data acquisition and processing. Their research focused on monitoring vital parameters such as heart rate and oxygen saturation. The system demonstrated improved processing performance and real-time monitoring capabilities, although the use of FPGA increased the hardware complexity and system cost.

In 2025, B. Baraeinejad, M. Shokouhifar, and A. Jalali in the paper “Design and Implementation of an IoT-Based Respiratory Motion Sensor” introduced a respiratory monitoring system capable of detecting breathing patterns using IoT-enabled sensors. Their research focused on accurate respiratory motion detection for healthcare applications. The study highlighted improved sensing accuracy and reliable signal acquisition; however, the system mainly concentrated on respiratory monitoring without integrating other physiological parameters.

In 2025, G. Georgieva-Tsaneva, V. Gancheva, I. Ruskova, and D. Cheshmedzhiev in the paper “Healthcare Monitoring Using an Internet of Things-Based Cardio System” presented a cardio monitoring framework that utilizes IoT technology for continuous tracking of heart-related parameters. Their work emphasized remote monitoring capabilities and efficient communication between wearable sensors and healthcare platforms.

In 2025, G. Georgieva-Tsaneva, V. Gancheva, and I. Ruskova in the paper “HRV Analysis in IoT-Based Cardiac Monitoring Systems” explored the use of heart rate variability (HRV) analysis in IoT-enabled cardiac monitoring systems. Their research highlighted the significance of HRV analysis for identifying cardiovascular abnormalities and predicting potential health risks. The study showed that HRV-based monitoring can improve the accuracy of cardiac health analysis.

In 2025, A. G. Mohapatra, S. Nayak, P. Das, and R. Mishra in the paper “IoT-Driven Remote Health Monitoring System” proposed a remote healthcare monitoring system that uses wearable sensors and wireless communication technologies to track patient health data continuously. Their work emphasized improved healthcare accessibility through remote monitoring and real-time data transmission to cloud platforms.

In 2025, A. Mohapatra, S. Nayak, and P. Das in the paper “Sensor Fusion in IoT Healthcare Systems” investigated the use of sensor fusion techniques for improving the accuracy and reliability of health monitoring systems. Their research demonstrated that combining multiple sensors can enhance signal quality and provide more accurate physiological measurements in IoT-based healthcare applications.

In 2025, S. Nayab, M. Arif, A. Khan, and F. Iqbal in the paper “IoT Framework for Cardiovascular Monitoring” proposed a cardiovascular monitoring framework that integrates wearable sensors with IoT networks. Their research focused on continuous heart monitoring and remote healthcare management. The system showed improved communication efficiency and real-time monitoring capabilities.

In 2025, J. P. Jose, R. Varghese, M. Thomas, and S. Philip in the paper “IoT Enabled Health Monitoring System Using Optimization and Deep Learning” developed an intelligent health monitoring system integrating deep learning techniques with IoT devices. Their work emphasized predictive healthcare analysis and disease detection using advanced machine learning algorithms. However, the implementation required higher computational resources.

In 2024, J. B. Harez, M. S. Ahmed, A. K. Hassan, and R. K. Ibrahim in the paper “IoT-Driven System for Continuous Monitoring of Heart Disease Patients Post-Surgery” presented a monitoring system designed specifically for patients recovering from cardiac surgery. Their system enabled continuous monitoring of vital signs and provided remote access to patient data for healthcare professionals.

In 2024, K. Dheman, R. Sharma, A. Gupta, and P. Mehta in the paper “i-CardiAx: Wearable IoT-Driven System for Vital Sign Monitoring” proposed a wearable healthcare device capable of monitoring multiple physiological parameters. Their work focused on integrating wearable sensors with IoT platforms for real-time health monitoring and improved patient mobility.

In 2024, S. K. Paul, M. Hasan, R. Islam, and T. Rahman in the paper “IoT-Based Real-Time Healthcare Monitoring with Deep Learning” introduced a system combining IoT technology with deep learning models for real-time patient monitoring. Their research highlighted improved prediction accuracy and intelligent health data analysis.

In 2023, F. Ali, S. El-Sappagh, A. Ali, and K. Kwak in the paper “Cloud-Based IoT Health Monitoring System” proposed a healthcare monitoring system that integrates IoT devices with cloud computing platforms. Their system focused on secure data storage, remote accessibility, and continuous monitoring of patient health parameters.

In 2023, L. Chen, X. Zhang, Y. Liu, and Z. Wang in the paper “Remote Patient Monitoring Using IoT and AI” developed a system that integrates IoT sensors with artificial intelligence algorithms for analyzing patient health data. Their research demonstrated improved accuracy in health monitoring and emphasized the role of AI in predictive healthcare applications.

In 2023, S. Kumar, R. Singh, P. Verma, and A. Sharma in the paper “Smart IoT-Based Healthcare Monitoring System” presented a smart healthcare system that uses IoT sensors for monitoring vital parameters such as heart rate, temperature, and oxygen saturation. Their work emphasized low-cost implementation and improved accessibility for remote healthcare services.

In 2022, S. Abdulmalek, A. Nasr, M. A. Ahmed, and K. S. Khalifa in the paper “IoT-Based Healthcare Monitoring System: A Review” provided a comprehensive review of IoT healthcare monitoring technologies. Their research discussed various architectures, communication protocols, and sensor technologies used in healthcare systems.

In 2022, A. Gupta, R. Kumar, S. Jain, and P. Sharma in the paper “IoT-Based Smart Patient Monitoring System” proposed a patient monitoring system using IoT sensors to measure physiological parameters and transmit the data to cloud platforms for remote analysis.

In 2022, M. Khan, S. Rehman, A. Ullah, and T. Ali in the paper “IoT-Based Wearable Health Monitoring Devices” investigated wearable sensor technologies for healthcare monitoring. Their research highlighted the importance of wearable devices for continuous physiological monitoring and early disease detection.

In 2022, V. Patel, N. Shah, R. Mehta, and S. Desai in the paper “Smart Healthcare Monitoring Using IoT Sensors” proposed a healthcare monitoring system that integrates IoT sensors and wireless communication technologies to improve patient monitoring and healthcare accessibility.

2.1 SUMMARY OF LITERATURE

From the reviewed literature, it is evident that IoT technology has significantly transformed healthcare monitoring by enabling continuous data collection and remote patient observation. Most of the existing systems focus on monitoring basic physiological parameters such as heart rate, oxygen saturation, or respiratory signals using wearable sensors and wireless communication technologies.

Several studies have also explored the integration of artificial intelligence, deep learning, and sensor fusion techniques to improve the accuracy and reliability of health monitoring systems. To address these limitations, the proposed project focuses on developing an IoT-based remote health surveillance system. By combining these parameters, the system generates a Smart Health Score that provides a complete overview of the user’s health condition.

CHAPTER 3

EXISTING AND PROPOSED METHOD

3.1 EXISTING METHOD

The existing healthcare monitoring systems are primarily based on traditional approaches that require patients to visit hospitals or clinics for regular health check-ups. In these systems, vital parameters such as heart rate, blood pressure, respiratory rate, and oxygen saturation (SpO₂) are measured using standalone medical instruments like sphygmomanometers, pulse oximeters, and electrocardiogram (ECG) machines. These devices are generally operated by trained healthcare professionals, and the readings are recorded manually or stored in isolated systems. Although these methods provide accurate and reliable measurements, they are not suitable for continuous monitoring. Patients with chronic illnesses or critical conditions require frequent monitoring, which becomes inconvenient and time-consuming in such setups. Additionally, the absence of real-time monitoring makes it difficult to detect sudden abnormalities, thereby increasing the risk of delayed medical intervention.

In hospital-based monitoring systems, patients admitted to intensive care units (ICUs) or general wards are continuously monitored using advanced bedside monitoring devices. These systems can track multiple physiological parameters simultaneously and provide real-time data to healthcare providers. However, these monitoring systems are highly expensive and require sophisticated infrastructure, making them accessible only in well-equipped hospitals. Moreover, patients need to remain physically present in the hospital to benefit from such monitoring, which is not always feasible, especially for elderly individuals or patients with mobility issues. The lack of portability and dependency on hospital infrastructure limits the widespread usage of these systems for long-term health monitoring.

With technological advancements, wearable health monitoring devices such as smartwatches and fitness trackers have gained popularity. These devices are capable of measuring basic health parameters such as heart rate, physical activity, calorie consumption.

Designed for fitness and wellness purposes rather than clinical applications. The accuracy of these devices may not meet medical standards, and they are generally limited in terms of the number of parameters they can measure. For instance, most wearable devices do not provide reliable measurements of blood pressure, stress levels, or respiratory rate. Furthermore, these devices lack integration with professional healthcare systems, making it difficult for doctors to utilize the collected data for diagnosis and treatment. The absence of advanced data analysis and decision-making capabilities further reduces their effectiveness in critical healthcare scenarios.

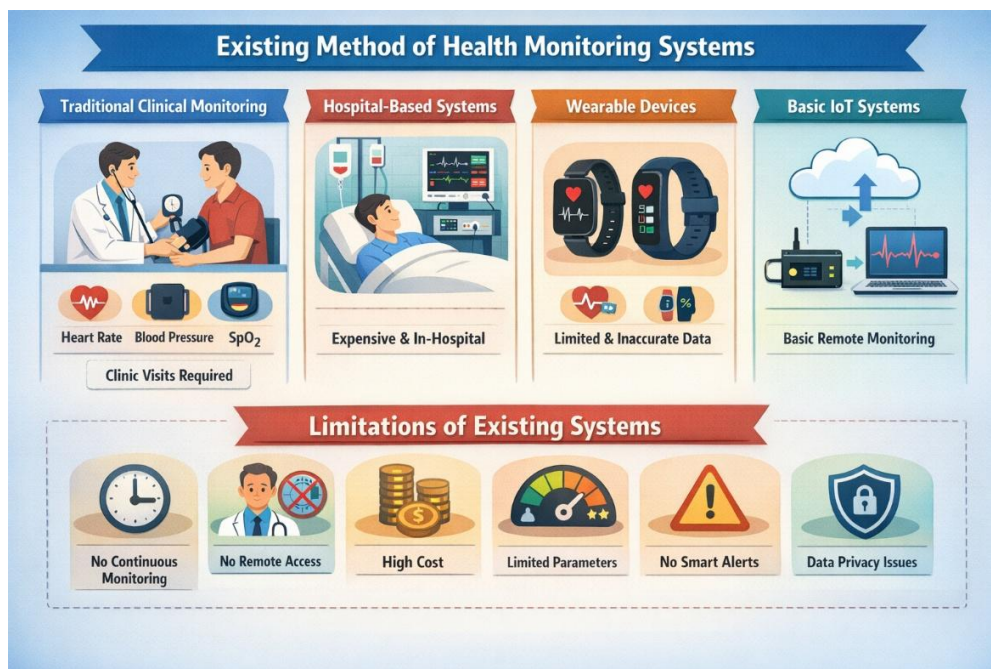


Figure 3.1 Existing Wearable Health Monitoring and Safety System

In recent years, basic Internet of Things (IoT)-based healthcare monitoring systems have been introduced to address some of the limitations of traditional and wearable devices. These systems use sensors to collect physiological data from patients and transmit the data to remote servers or cloud platforms using microcontrollers and communication technologies. This enables remote monitoring of patients and allows healthcare providers to access patient data from distant locations. Although IoT-based systems represent a significant improvement over traditional methods, many of the existing implementations are still limited in functionality. Most systems focus on monitoring only one or two parameters, such as heart rate, and do not provide a comprehensive view of the patient's health condition.

Another major limitation of existing IoT-based systems is the lack of intelligent data processing. While these systems can collect and transmit data, they often do not include advanced algorithms to analyze the data and generate meaningful insights. As a result, healthcare providers must manually interpret the data, which can be time-consuming and prone to errors. Additionally, these systems generally do not include a smart health scoring mechanism that combines multiple parameters into a single, easy-to-understand metric. This makes it difficult for patients and caregivers to quickly assess the overall health status.

Data security and privacy are also significant concerns in existing healthcare monitoring systems. Many IoT-based systems lack robust security mechanisms to protect sensitive patient data during transmission and storage. This increases the risk of data breaches and unauthorized access, which can have serious implications for patient confidentiality. Furthermore, issues related to system reliability and network dependency can affect the performance of these systems. In cases of poor internet connectivity, data transmission may be delayed or interrupted, reducing the effectiveness of real-time monitoring.

Another drawback of existing systems is the absence of effective alert mechanisms. Although some systems provide basic alerts when parameters exceed certain thresholds, they may not be reliable or timely enough to handle critical situations. In many cases, alerts are not integrated with emergency response systems, limiting their usefulness in saving lives during emergencies. Moreover, most systems do not support predictive analysis, which could help in identifying potential health risks before they become severe.

The lack of integration between different components of healthcare systems is another significant issue. Existing systems often operate in isolation, without proper coordination between sensors, data processing units, cloud platforms, and healthcare providers. This results in fragmented data and reduces the overall efficiency of healthcare monitoring. Additionally, many systems are not user-friendly and require technical expertise to operate, making them unsuitable for elderly patients or non-technical users. Cost is another important factor that limits the adoption of existing healthcare monitoring systems. Advanced hospital-based systems are expensive for a large portion of the population.

In many cases, alerts are not integrated with emergency response systems, limiting their usefulness in saving lives during emergencies. Even some IoT-based solutions may involve high initial setup costs and maintenance expenses. This creates a barrier for widespread implementation, especially in rural and low-resource settings where affordable healthcare solutions are most needed. Existing systems often operate in isolation, without proper coordination between sensors, data processing units, cloud platforms, and healthcare providers.

Overall, the existing methods of healthcare monitoring suffer from several limitations, including lack of continuous and real-time monitoring, dependency on hospital infrastructure, limited parameter measurement, absence of intelligent data analysis, security concerns, and high costs. These challenges highlight the need for an improved system that can provide comprehensive, real-time, and cost-effective health monitoring with advanced features such as remote accessibility, multi-parameter analysis, and smart health scoring. Such a system would significantly enhance patient care, enable early detection of health issues, and support the development of modern smart healthcare solutions.

3.2 PROPOSED METHOD

In recent years, basic Internet of Things (IoT)-based healthcare monitoring systems have been introduced to address some of the limitations of traditional and wearable devices. These systems use sensors to collect physiological data from patients and transmit the data to remote servers or cloud platforms using microcontrollers and communication technologies. This enables remote monitoring of patients and allows healthcare providers to access patient data from distant locations.

Although IoT-based systems represent a significant improvement over traditional methods, many of the existing implementations are still limited in functionality. Most systems focus on monitoring only one or two parameters, such as heart rate or temperature, and do not provide a comprehensive view of the patient's health condition. Another major limitation of existing IoT-based systems is the lack of intelligent data processing. While these systems can collect and transmit data, they often do not include advanced algorithms to analyze the data and generate meaningful insights.

As a result, healthcare providers must manually interpret the data, which can be time-consuming and prone to errors. Additionally, these systems generally do not include a smart health scoring mechanism that combines multiple parameters into a single, easy-to-understand metric. This makes it difficult for patients and caregivers to quickly assess the overall health status.

Data security and privacy are also significant concerns in existing healthcare monitoring systems. Many IoT-based systems lack robust security mechanisms to protect sensitive patient data during transmission and storage. This increases the risk of data breaches and unauthorized access, which can have serious implications for patient confidentiality. Furthermore, issues related to system reliability and network dependency can affect the performance of these systems. In cases of poor internet connectivity, data transmission may be delayed or interrupted, reducing the effectiveness of real-time monitoring.

Another drawback of existing systems is the absence of effective alert mechanisms. Although some systems provide basic alerts when parameters exceed certain thresholds, they may not be reliable or timely enough to handle critical situations. In many cases, alerts are not integrated with emergency response systems, limiting their usefulness in saving lives during emergencies. Moreover, most systems do not support predictive analysis, which could help in identifying potential health risks before they become severe.

The lack of integration between different components of healthcare systems is another significant issue. Existing systems often operate in isolation, without proper coordination between sensors, data processing units, cloud platforms, and healthcare providers. This results in fragmented data and reduces the overall efficiency of healthcare monitoring. Additionally, many systems are not user-friendly and require technical expertise to operate, making them unsuitable for elderly patients or non-technical users.

Cost is another important factor that limits the adoption of existing healthcare monitoring systems. Advanced hospital-based systems are expensive and not affordable for a large portion of the population. Even some IoT-based solutions may involve high initial setup costs and maintenance expenses. Although some systems provide basic alerts when parameters exceed certain thresholds, they may not be enough to handle critical situations.

Overall, the existing methods of healthcare monitoring suffer from several limitations, including lack of continuous and real-time monitoring, dependency on hospital infrastructure, limited parameter measurement, absence of intelligent data analysis, security concerns, and high costs. These challenges highlight the need for an improved system that can provide comprehensive, real-time, and cost-effective health monitoring with advanced features such as remote accessibility, multi-parameter analysis, and smart health scoring. Such a system would significantly enhance patient care, enable early detection of health issues, and support the development of modern smart healthcare solutions.

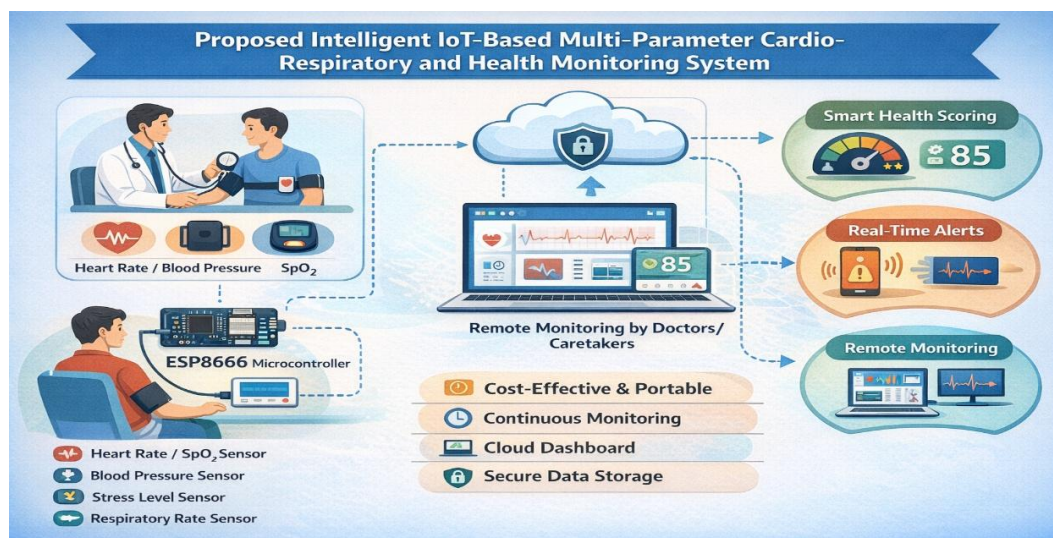


Figure 3.2 Proposed Method of IoT-Based Multi-Parameter Cardio Respiratory and Health Monitoring with Smart Health Scoring

The proposed system also incorporates an intelligent alert mechanism to enhance patient safety. Threshold values are predefined for each health parameter, and whenever the measured values exceed these limits, the system automatically generates alerts. This feature plays a critical role in preventing severe health complications and ensures quick response during emergencies.

A unique feature of the proposed system is the implementation of a smart health scoring mechanism. This mechanism analyzes multiple physiological parameters and combines them into a single health score that represents the overall condition of the patient. The health score is calculated using predefined algorithms that consider factors such as normal ranges, deviations, and trends of each parameter.

This simplifies complex medical data into an easy-to-understand format, allowing patients and caregivers to quickly assess health status without requiring extensive medical knowledge. It also assists healthcare professionals in making faster and more accurate diagnoses. These alerts are sent to doctors, caregivers, or family members through notifications, enabling immediate attention and timely medical intervention.

The system is designed to be highly portable, user-friendly, and cost-effective, making it suitable for both hospital and home environments. Unlike traditional systems, it does not require bulky equipment or continuous supervision by medical staff. This is especially beneficial for elderly individuals, physically challenged patients, and those living in remote areas with limited access to healthcare facilities.

In addition, the proposed system supports scalability and flexibility, allowing the integration of additional sensors and features in the future. The modular design of the system enables easy upgrades and customization based on specific healthcare requirements. For example, additional parameters such as body temperature or ECG monitoring can be incorporated to further enhance the system's capabilities.

Data security and privacy are also considered in the proposed system. Secure communication protocols and authentication mechanisms can be implemented to protect sensitive patient data during transmission and storage. This ensures that only authorized users can access the data, maintaining confidentiality and trust in the system. Furthermore, the proposed system reduces the workload on healthcare professionals by automating data collection, analysis, and reporting processes. It enables continuous monitoring without the need for manual intervention, thereby improving efficiency and reducing human errors.

Overall, the proposed Intelligent IoT-Based Multi-Parameter Cardio-Respiratory and Health Monitoring System with Smart Health Scoring provides a comprehensive and reliable solution for modern healthcare challenges. The system also promotes preventive healthcare by identifying potential health risks at an early stage, allowing timely treatment and reducing the chances of complications. By complicating the needs of user like things of

remote accessibility, intelligent analysis, and cost-effective design, the system significantly enhances patient care and safety. It bridges the gap between patients and healthcare providers, enabling efficient and smart healthcare management. This system represents a significant step toward the development of advanced digital healthcare solutions and supports the vision of accessible and continuous health monitoring for all individuals.

3.3 BLOCK DIAGRAM

The block diagram represents the working architecture of the “Intelligent IoT-Based Multi-Parameter Cardio-Respiratory and Health Monitoring System with Smart Health Scoring System.” This system is designed to continuously monitor various physiological parameters of a patient and transmit the collected data to a cloud platform for real-time analysis and remote access. The overall system consists of input sensors, a processing unit (ESP8266 microcontroller), communication modules, and a cloud-based monitoring dashboard. Each block in the diagram plays a vital role in ensuring accurate data acquisition, processing, and transmission.

The first section of the block diagram consists of multiple biosensors that are responsible for collecting real-time physiological data from the human body. These sensors include the heart rate sensor, SpO2 sensor, blood pressure sensor, stress detection sensor, and respiratory rate sensor. The heart rate sensor measures the number of heartbeats per minute by detecting the pulse of blood flow through the body. This parameter is essential for identifying abnormal heart conditions such as tachycardia or bradycardia. The SpO2 sensor measures the oxygen saturation level in the blood, which indicates how effectively oxygen is being transported to different parts of the body.

The blood pressure sensor is used to estimate the systolic and diastolic pressure levels in the body. Monitoring blood pressure is crucial for detecting hypertension or hypotension, which can lead to serious health complications if left untreated. This parameter is particularly important for patients with respiratory issues or conditions such as asthma and chronic obstructive pulmonary disease. This system is designed to continuously monitor various physiological.

The stress detection sensor, often implemented using galvanic skin response (GSR), measures the electrical conductivity of the skin, which varies with sweat gland activity. This helps in identifying stress levels and emotional conditions of the patient. Additionally, the respiratory rate sensor measures the number of breaths taken per minute, which is a key indicator of lung function and overall respiratory health. All these sensors work together to provide a comprehensive set of health data.

The outputs from these sensors are fed into the central processing unit, which is the ESP8266 microcontroller. The ESP8266 is a low-cost, high-performance microcontroller with built-in Wi-Fi capabilities, making it ideal for IoT-based applications. It acts as the brain of the system by collecting data from all connected sensors and performing initial processing. The microcontroller converts the analog signals received from the sensors into digital data using an internal analog-to-digital converter (ADC). It then processes this data to ensure accuracy and prepares it for transmission.

One of the key advantages of using the ESP8266 is its ability to connect directly to a wireless network. After processing the sensor data, the microcontroller transmits the information to the cloud through the internet using Wi-Fi communication. The data transmission is carried out in real-time, ensuring that the latest health information is always available for monitoring.

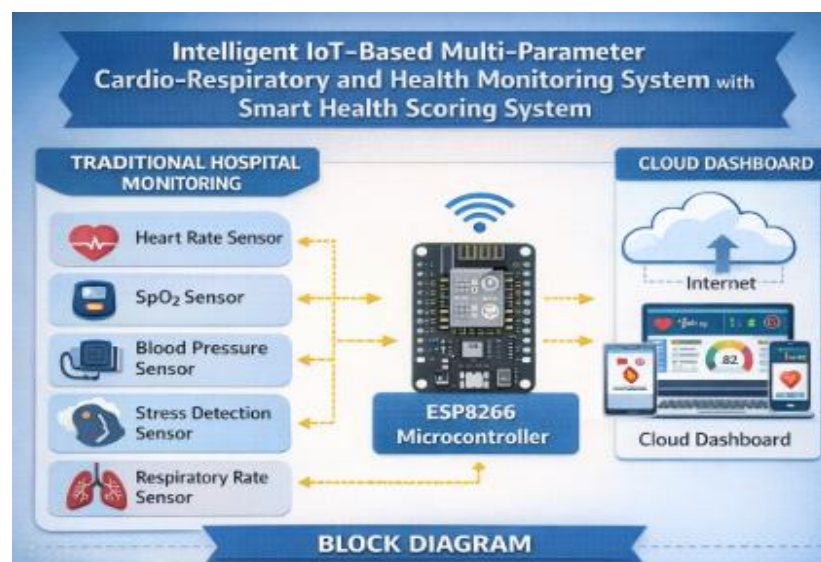


Figure 3.3 Block Diagram of Proposed Smart Health Scoring System

The next block in the diagram represents the cloud platform, which acts as a centralized data storage and management system. The cloud receives the data sent by the ESP8266 and stores it securely for further analysis. This data can include real-time readings as well as historical records, enabling long-term monitoring of the patient's health. Cloud integration is a crucial aspect of the system, as it allows healthcare professionals to access patient data remotely from any location. It also supports scalability, meaning that multiple patients can be monitored simultaneously using the same platform.

The cloud platform is connected to a user interface, typically a web-based dashboard or mobile application, which is shown in the block diagram as the cloud dashboard. This dashboard provides a visual representation of the patient's health parameters in the form of graphs, charts, and numerical values. Doctors, caregivers, and patients can log in to the dashboard to view real-time data and track changes in health conditions over time. The dashboard also enhances usability by presenting complex data in an easy-to-understand format.

Another important feature of the system, though not explicitly shown as a separate block, is the alert mechanism. The system is programmed with predefined threshold values for each health parameter. If any parameter exceeds or falls below the normal range, the system automatically generates alerts. The integration of all these components results in a highly efficient and intelligent health monitoring system. The sensors continuously collect data, the ESP8266 processes and transmits it, the cloud stores and analyzes it, and the dashboard displays it for users. This seamless flow of information ensures real-time monitoring and effective decision-making. Unlike traditional systems, this setup does not require the patient to be physically present in a hospital, thereby reducing the burden on healthcare facilities and improving convenience for patients.

Furthermore, the system supports continuous monitoring, which is essential for detecting sudden changes in health conditions. It is particularly beneficial for elderly individuals, patients with chronic diseases, and those recovering from surgeries. The portability and ease of use of the system make it suitable for home-based healthcare, allowing patients to monitor their health comfortably without frequent hospital visits.

In conclusion, the block diagram illustrates a well-integrated IoT-based healthcare monitoring system that combines multiple sensors, a powerful microcontroller, and cloud technology to provide real-time, accurate, and remote health monitoring. Each component in the system plays a significant role in ensuring the reliability and effectiveness of the overall solution. By enabling continuous monitoring, remote access, and timely alerts, this system significantly enhances patient care and represents a major advancement in modern healthcare technology.

3.4 CIRCUIT DIAGRAM

The circuit diagram of the “Intelligent IoT-Based Multi-Parameter Cardio-Respiratory and Health Monitoring System with Smart Health Scoring System” illustrates the hardware connections between various sensors, the ESP8266 microcontroller, power supply components, and communication interfaces. This circuit is designed to acquire physiological data from the human body, process it, and transmit it to the cloud for real-time monitoring. Each component in the circuit plays a crucial role in ensuring accurate data collection, efficient processing, and reliable communication.

At the core of the circuit is the ESP8266 microcontroller, which acts as the central processing and communication unit. The ESP8266 is selected due to its low cost, compact size, and built-in Wi-Fi capability, making it highly suitable for IoT-based applications. It contains General Purpose Input/Output (GPIO) pins that are used to interface with different sensors. The microcontroller operates at a voltage level of 3.3V, and therefore, all connected components must be compatible with this voltage or use appropriate voltage regulation techniques. The ESP8266 collects analog and digital signals from the sensors, processes the data, and transmits it to the cloud via a wireless network.

The heart rate and SpO2 sensor is one of the primary components connected to the microcontroller. Typically, sensors like MAX30100 or MAX30102 are used, which operate based on photoplethysmography (PPG) principles. These sensors are connected to the ESP8266 using I2C communication protocol, where the Serial Data Line (SDA) and Serial Clock Line (SCL) pins are interfaced with the corresponding GPIO pins of the microcontroller. The sensor also requires power supply connections, usually VCC and GND, to function properly.

The blood pressure estimation module is another important part of the circuit. In systems, blood pressure is either measured using a digital BP sensor module or estimated using pulse waveform analysis. The output of this sensor is typically analog in nature and is connected to the Analog-to-Digital Converter (ADC) pin of the ESP8266. Since the ESP8266 has limited ADC capability, proper signal conditioning may be required to ensure accurate readings. The sensor must also be powered with the correct voltage, and in some cases, additional components like resistors or amplifiers are used to stabilize the signal.

The stress detection sensor, commonly implemented using a Galvanic Skin Response (GSR) sensor, measures the electrical conductivity of the skin. This sensor detects variations in sweat gland activity, which are associated with stress levels. The GSR sensor produces an analog output signal, which is connected to the ADC pin of the ESP8266. The circuit may include resistors to control current flow and improve measurement accuracy. Proper grounding is essential in this part of the circuit to avoid noise and interference.

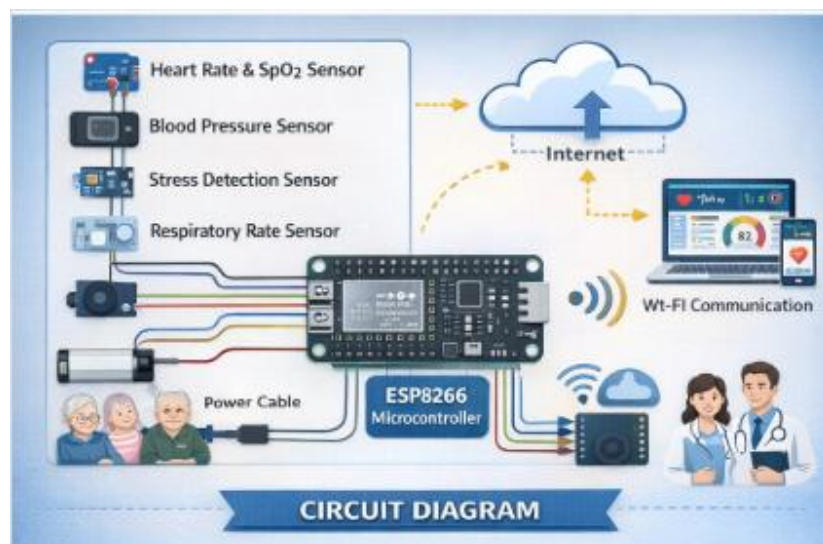


Figure 3.4 Circuit Diagram of Proposed Smart Health Scoring System

The respiratory rate sensor is used to measure the breathing pattern of the patient. This sensor can be implemented using different methods such as temperature sensors, pressure sensors, or chest movement detection modules. In the circuit, the sensor is connected to the microcontroller either through analog input pins or digital GPIO pins,

depending on these type of sensor used. The signal from the respiratory sensor is processed by the microcontroller to calculate the number of breaths per minute. The sensor must also be powered with the correct voltage, and in some cases, additional components like resistors or amplifiers are used to stabilize the signal. This seamless flow of information ensures real-time monitoring and effective decision-making.

Power supply is a critical component of the circuit design. The entire system is powered using a regulated power source, which may include a battery or an external adapter. Since the ESP8266 operates at 3.3V, a voltage regulator is used to step down higher input voltages (such as 5V) to a stable 3.3V output. Components such as capacitors are included in the circuit to filter noise and maintain a stable voltage supply. Proper power management ensures the reliability and longevity of the system.

All sensors and components share a common ground (GND), which is essential for proper circuit operation. Without a common reference point, the signals may become unstable or inaccurate. Additionally, pull-up resistors are often used in I2C communication lines (SDA and SCL) to maintain proper signal levels and ensure reliable communication between the microcontroller and sensors.

The communication aspect of the circuit is handled by the built-in Wi-Fi module of the ESP8266. Once the sensor data is processed, the microcontroller sends the data to a cloud platform using internet connectivity. This wireless communication eliminates the need for wired data transfer and enables remote monitoring of patient health. The circuit is designed to ensure continuous data transmission without interruptions, provided that a stable internet connection is available.

Another important consideration in the circuit design is noise reduction and signal accuracy. Since biomedical signals are often very weak, the circuit must be designed carefully to minimize electrical noise and interference. This can be achieved by proper placement of components, use of shielding techniques, and inclusion of filtering components such as capacitors and resistors. Ensuring clean and stable signals is essential for accurate health monitoring.

The overall circuit is compact and designed for portability, allowing it to be used in home-based healthcare environments. The use of integrated sensors and a single microcontroller reduces the complexity of the circuit and makes it easier to assemble and maintain. Additionally, the modular nature of the design allows for future expansion, such as adding more sensors or upgrading components.

In conclusion, the circuit diagram represents a well-organized integration of sensors, microcontroller, power supply, and communication modules. The ESP8266 acts as the central unit that connects all components and facilitates real-time data transmission to the cloud. Each sensor is carefully interfaced to ensure accurate data collection, while the power supply and supporting components maintain system stability. The design emphasizes efficiency, reliability, and scalability, making it suitable for modern IoT-based healthcare applications. This circuit plays a crucial role in enabling continuous, real-time, and remote health monitoring, ultimately improving patient care and safety.

3.5 WORKING PRINCIPLE AND IMPLEMENTATION

The working principle of the “Intelligent IoT-Based Multi-Parameter Cardio-Respiratory and Health Monitoring System with Smart Health Scoring System” is based on continuous acquisition of physiological signals from the human body, real-time processing using a microcontroller, and wireless transmission of the data to a cloud platform for monitoring and analysis. The system operates by integrating multiple biosensors with an ESP8266 microcontroller, which acts as the central unit for data collection, processing, and communication. The overall operation begins with sensing, followed by signal conditioning, data processing, wireless transmission, cloud storage, and finally visualization through a user interface.

Initially, various sensors are attached to the patient’s body to measure important health parameters such as heart rate, oxygen saturation (SpO₂), blood pressure estimation, stress level, and respiratory rate. The heart rate and SpO₂ sensor works on the principle of photoplethysmography, where light is emitted through the skin and the variation in light absorption is used to determine pulse rate and oxygen levels in the blood.

The stress detection sensor measures the skin conductivity using galvanic skin response, which varies based on sweat gland activity and emotional state. The respiratory rate is measured by detecting breathing patterns through suitable sensors, while blood pressure is estimated using sensor-based techniques or derived from pulse characteristics. These sensors continuously generate electrical signals corresponding to physiological activities.

The signals obtained from the sensors are then fed into the ESP8266 microcontroller. Since some of these signals are analog in nature, they are converted into digital signals using an internal analog-to-digital converter (ADC). The microcontroller processes the incoming data by filtering noise, removing unwanted fluctuations, and calculating meaningful values such as beats per minute, oxygen percentage, breaths per minute, and stress levels. This step ensures that the data is accurate and suitable for further analysis. The processed data is then formatted into a structured form for transmission.

Once the data processing is completed, the ESP8266 uses its built-in Wi-Fi module to connect to a wireless network and transmit the data to a cloud server. This communication happens in real time, allowing continuous monitoring of the patient's health condition. The cloud platform acts as a centralized system where all the data is stored securely. It also performs additional processing, such as maintaining historical records, analyzing trends, and supporting data visualization. This ensures that both real-time and past data are available for medical evaluation.

A key feature of the working principle is the smart health scoring mechanism. The system evaluates multiple health parameters simultaneously and assigns a health score based on predefined thresholds and conditions. For example, if all parameters are within normal range, the health score will be high, indicating a stable condition. If one or more parameters deviate from normal values, the score decreases, indicating potential health risks. This communication happens in real time, allowing continuous monitoring of the patient's health condition. This simplifies complex medical data into an easily understandable format, enabling quick decision-making by healthcare professionals.

The system also incorporates an alert mechanism as part of its working principle. Whenever any health parameter crosses its predefined safe limit, the system automatically generates alerts and notifications. These alerts are sent to doctors, caregivers, or family members through the cloud interface, ensuring immediate attention. This feature is particularly useful in emergency situations, where early detection can prevent serious complications.

In terms of implementation, the system is developed by integrating hardware and software components in a systematic manner. The hardware setup includes sensors, ESP8266 microcontroller, power supply, and connecting wires. Each sensor is interfaced with the microcontroller using appropriate communication protocols such as I2C or analog input pins. Proper connections are ensured to maintain signal accuracy and system stability. A regulated power supply is used to provide consistent voltage to all components.

On the software side, the ESP8266 is programmed using embedded C or Arduino IDE to read sensor data, process it, and send it to the cloud. Libraries specific to each sensor are used to simplify communication and data acquisition. The microcontroller code includes functions for initializing sensors, reading data, performing calculations, and handling Wi-Fi communication. The cloud platform is configured to receive and store the transmitted data, and a dashboard is designed to display the data in graphical and numerical formats.

The implementation also involves testing and calibration of sensors to ensure accurate measurements. Each sensor is calibrated according to standard values to minimize errors. The system is tested under different conditions to verify its performance, reliability, and response time. Any discrepancies in readings are corrected through software adjustments or hardware modifications.

Furthermore, the system is designed to be portable and user-friendly, making it suitable for real-world applications. The compact design allows patients to use the system comfortably at home. The use of wireless communication eliminates the need for complex wiring, and the cloud-based interface ensures easy access to data from anywhere. This makes the system highly efficient for continuous health monitoring.

In conclusion, the working principle of the proposed system involves a seamless integration of sensing, processing, communication, and analysis to provide real-time health monitoring. The implementation combines reliable hardware components with efficient software algorithms to ensure accurate data collection and transmission. The addition of smart health scoring and alert mechanisms enhances the functionality of the system, making it a comprehensive solution for modern healthcare needs. This system not only improves patient safety but also supports remote healthcare management and early detection of health issues.

3.6 ADVANTAGES

The proposed “Intelligent IoT-Based Multi-Parameter Cardio-Respiratory and Health Monitoring System with Smart Health Scoring” offers numerous advantages over traditional healthcare monitoring methods. One of the most significant advantages is real-time continuous monitoring of multiple physiological parameters. Unlike conventional systems that require periodic hospital visits this continuous monitoring helps in early detection of abnormalities and ensures timely medical intervention, thereby reducing the risk of severe health complications.

Another major advantage is remote accessibility. The integration of IoT technology allows patient data to be transmitted to a cloud platform, where it can be accessed by doctors, caregivers, and family members from any location. This feature eliminates the need for frequent hospital visits and enables efficient remote patient monitoring. It is particularly beneficial for patients living in remote or rural areas where access to healthcare facilities is limited. Doctors can monitor patients in real time and provide guidance without requiring physical presence, improving healthcare efficiency.

The system is also cost-effective compared to traditional hospital monitoring systems. Conventional systems require expensive equipment and infrastructure, making them inaccessible to many patients. In contrast, the proposed system uses low-cost components such as sensors and the ESP8266 microcontroller, making it affordable and suitable for large-scale deployment. This affordability ensures that advanced healthcare monitoring is accessible to a wider population.

Another key advantage is the multi-parameter monitoring capability. Unlike wearable fitness devices that measure only basic parameters, this system integrates multiple sensors to provide a comprehensive view of the patient's health. Monitoring multiple parameters simultaneously improves accuracy in diagnosis and helps in better understanding of the patient's overall condition. This holistic approach enhances the effectiveness of healthcare monitoring.

The introduction of a smart health scoring system is a unique advantage of the proposed system. It combines different health parameters into a single score, making it easier for users to understand their health status. This feature simplifies complex medical data and allows both patients and healthcare professionals to quickly assess the condition. It also helps in tracking health improvements or deterioration over time.

The system includes an automatic alert mechanism, which enhances patient safety. Whenever any parameter exceeds its normal range, the system generates alerts and notifications to doctors or caregivers. This ensures immediate response during critical situations and helps prevent emergencies. Early warning systems play a vital role in saving lives, especially for patients with chronic conditions.

Another important advantage is the portability and ease of use of the system. The compact design allows patients to use it comfortably at home without requiring technical expertise. The user-friendly interface and wireless connectivity make it convenient for daily use. This is particularly beneficial for elderly individuals and patients with mobility issues.

The system also provides data storage and historical analysis through cloud integration. All collected data is stored securely in the cloud, allowing long-term tracking of health conditions. Doctors can analyze trends and patterns in patient data, which helps in better diagnosis and treatment planning. This feature eliminates the need for frequent hospital visits and enables efficient remote patient monitoring. It is particularly beneficial for patients living in remote or rural areas where access to healthcare facilities is limited.

This feature supports preventive healthcare by identifying potential risks before they become serious. Additionally, the system ensures scalability and flexibility. New sensors and features can be easily added to the existing setup, making it adaptable to future technological advancements. This flexibility allows the system to evolve according to changing healthcare needs.

Finally, the proposed system improves overall healthcare efficiency by reducing the workload on healthcare professionals. Automated data collection, processing, and reporting minimize manual effort and reduce human errors. This leads to more accurate monitoring and better patient care.

3.7 APPLICATIONS

The proposed IoT-based health monitoring system has a wide range of applications in various healthcare and real-world scenarios. One of the primary applications is in remote patient monitoring. The system allows doctors to monitor patients from a distance using cloud-based dashboards. This is especially useful for patients with chronic diseases such as heart conditions, respiratory disorders, and hypertension, who require continuous monitoring. Remote monitoring reduces hospital visits and improves patient convenience.

Another important application is in home healthcare systems. The system enables patients to monitor their health from the comfort of their homes. This is highly beneficial for elderly individuals, physically challenged patients, and those recovering from surgeries. Home-based monitoring improves quality of life and reduces dependency on hospital infrastructure. It also helps in early detection of health issues, preventing complications.

The system can also be used in hospital and clinical environments as a supplementary monitoring tool. It can assist healthcare professionals in tracking multiple patients simultaneously and maintaining digital health records. This improves efficiency in hospitals and reduces the burden on medical staff. The system can also be integrated with existing hospital management systems for better coordination. Additionally, the system ensures scalability and flexibility. New sensors and features can be easily added to the existing setup, making it adaptable to future technological advancements.

Another significant application is in emergency healthcare services. The real-time monitoring and alert mechanism enable quick detection of critical conditions such as sudden drop in oxygen levels or abnormal heart rate. Alerts can be sent immediately to medical personnel, allowing faster response and potentially saving lives. This makes the system suitable for use in ambulances and emergency care units.

The proposed system is also useful in telemedicine applications. With the increasing demand for virtual healthcare services, this system enables doctors to diagnose and monitor patients remotely. Patients can share their health data with doctors through the cloud platform, facilitating online consultations. This reduces the need for physical visits and improves accessibility to healthcare services.

Another application is in fitness and wellness monitoring. Although the system is designed for medical purposes, it can also be used by individuals to track their health parameters and maintain a healthy lifestyle. Monitoring stress levels, heart rate, and oxygen levels helps users stay aware of their health condition and take preventive measures.

The system can also be applied in rural healthcare development. In many rural areas, access to healthcare facilities is limited. The proposed system provides an affordable and efficient solution for monitoring patient health remotely. Healthcare workers can use the system to collect data and share it with doctors located in urban areas, improving healthcare delivery in underserved regions.

Furthermore, the system has applications in research and data analysis. The continuous collection of health data can be used for medical research and analysis. Researchers can study patterns, identify trends, and develop better treatment methods based on real-time data. This contributes to advancements in medical science and technology. The system is also suitable for use in workplace health monitoring, especially in industries where workers are exposed to stressful or hazardous environments. This is highly beneficial for elderly individuals, physically challenged patients, and those recovering from surgeries. Continuous monitoring of vital parameters can help in ensuring worker safety and preventing health-related incidents. This improves efficiency in hospitals and reduces the burden on medical staff.

The smart health scoring system is a unique advantage of the proposed system. It combines different health parameters into a single score, making it easier for users to understand their health status. This feature simplifies complex medical data and allows both patients and healthcare professionals to quickly assess the condition. It also helps in tracking health improvements or deterioration over time.

The system includes an automatic alert mechanism, which enhances patient safety. Whenever any parameter exceeds its normal range, the system generates alerts and notifications to doctors or caregivers. This ensures immediate response during critical situations and helps prevent emergencies. Early warning systems play a vital role in saving lives, especially for patients with chronic conditions.

Another important advantage is the portability and ease of use of the system. The compact design allows patients to use it comfortably at home without requiring technical expertise. The user-friendly interface and wireless connectivity make it convenient for daily use. . Monitoring multiple parameters simultaneously improves accuracy in diagnosis and helps in better understanding of the patient's overall condition. This contributes to advancements in medical science and technology. The system is also suitable for use in workplace health monitoring, especially in industries where workers are exposed to stressful or hazardous environments. This holistic approach enhances the effectiveness of healthcare monitoring. This is particularly beneficial for elderly individuals and patients with mobility issues.

In conclusion, the proposed system offers numerous advantages such as real-time monitoring, remote accessibility, cost-effectiveness, multi-parameter analysis, and smart health scoring. Its wide range of applications in healthcare, telemedicine, home care, emergency services, and research makes it a highly valuable solution for modern healthcare challenges. By improving accessibility, efficiency, and reliability, the system plays a crucial role in advancing smart healthcare technologies and enhancing patient well-being.

CHAPTER 4

HARDWARE AND SOFTWARE REQUIREMENTS

4.1 HARDWARE REQUIREMENTS

4.1.1 ESP32

NodeMCU (ESP8266) is the core microcontroller unit used in this system. It plays a vital role in controlling the entire operation of the device. The module comes with an inbuilt Wi-Fi feature, making it highly suitable for IoT-based applications. The ESP32 module used in the system is shown in Figure. 4.1.1.



Figure 4.1.1 ESP32 Microcontroller

The NodeMCU reads data from the pulse oximeter sensor and processes it using embedded programming. It converts analog signals into digital values and performs necessary calculations to determine heart rate and SpO₂ levels. After processing, it transmits the data to the cloud platform through Wi-Fi connectivity.

One of the major advantages of NodeMCU is its low cost and ease of programming using the Arduino IDE. It also consumes less power, making it ideal for portable healthcare systems. These variations are detected by the sensor and converted into electrical signals, which are then processed to calculate heart rate and oxygen saturation. Its compact size and integration capabilities make it suitable for real-time monitoring applications.

Additionally, the NodeMCU is known for its low power consumption and cost-effectiveness, making it ideal for portable healthcare systems. Its compatibility with Arduino IDE simplifies programming, even for beginners. Due to its reliability, compact size, and efficient performance, it plays a critical role in ensuring smooth system operation.

4.1.2 PULSE OXIMETER SENSOR (MAX30100 / MAX30102)



Figure 4.1.2 Pulse Oximeter Sensor

The MAX30100/MAX30102 pulse oximeter sensor is a key component responsible for measuring vital physiological parameters. It is designed to measure heart rate and blood oxygen saturation (SpO_2) non-invasively, making it highly suitable for continuous health monitoring systems. The MAX30102 sensor module used in the system is shown in Figure 4.1.2.

The sensor operates based on the principle of photoplethysmography (PPG), where it emits red and infrared light through the skin (typically a fingertip). The amount of light absorbed by the blood varies depending on oxygen levels and blood flow. These variations are detected by the sensor and converted into electrical signals, which are then processed to calculate heart rate and oxygen saturation. The MAX30102 version offers improved accuracy, better noise reduction, and enhanced performance compared to earlier models. It includes built-in filtering and signal processing features, which reduce interference from motion and ambient light. This ensures reliable readings even in real-time conditions.

The sensor is compact, energy-efficient, and easy to interface with microcontrollers using I2C communication. Its ability to provide continuous and accurate data makes it essential for detecting abnormal health conditions such as low oxygen levels or irregular heartbeats. This plays a vital role in early diagnosis and timely medical intervention.

4.1.3 OLED DISPLAY

The 0.96-inch OLED display is used as the primary output interface for the system. It provides real-time visualization of the measured health parameters, allowing users to monitor their condition instantly without relying on external devices. The EMG sensor module used in the system is shown in Figure. 4.1.3.

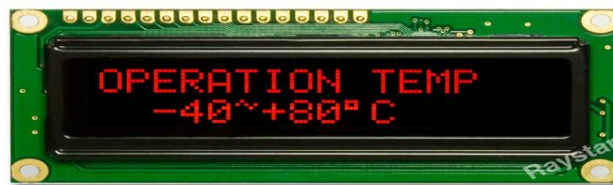


Figure 4.1.3 EMG Sensor Module with Electrodes

OLED (Organic Light Emitting Diode) technology offers several advantages over traditional LCD displays. It provides high contrast, faster response time, and better visibility even under low lighting conditions. Each pixel in an OLED display emits its own light, eliminating the need for a backlight and reducing power consumption.

In this system, the OLED display is interfaced with the NodeMCU using I2C communication. It continuously updates and displays values such as heart rate and SpO₂. The compact size of the display makes it ideal for portable and wearable applications. The display enhances user interaction by providing immediate feedback, which is particularly useful in emergency situations. It ensures that users can quickly check their health status without accessing cloud platforms or mobile applications.

4.1.4 POWER SUPPLY (5V USB)

The power supply unit is responsible for delivering stable and reliable power to all components in the system. The system can be powered using a 5V USB source or a rechargeable battery, depending on the application. A stable voltage supply is crucial for accurate sensor readings and proper functioning of the microcontroller. Any fluctuations in power can lead to incorrect data or system malfunction. Therefore, regulated power sources are preferred.

Using a battery as a power source enhances portability, allowing the system to be used in remote areas or for wearable applications. Rechargeable lithium-ion batteries are commonly used due to their high energy density and long lifespan. In this system, the OLED display is interfaced with the ESP32 microcontroller using the I2C communication protocol, which reduces wiring complexity by using only two main data lines (SDA and SCL). Efficient power management is also important in IoT systems. The NodeMCU and sensor are designed to consume minimal power, which helps in extending battery life. This ensures continuous operation of the system for long durations without frequent recharging.

4.2 SOFTWARE REQUIREMENTS

The software requirements form a vital part of the proposed IoT-based cardio-respiratory monitoring system, as they are responsible for controlling the overall operation and enabling intelligent functionality. While the hardware components are used to sense and collect physiological data, it is the software that processes this data, converts it into meaningful information, and ensures its proper transmission and visualization. Without an efficient software framework, the system would not be able to perform real-time monitoring or provide accurate results.

In this system, the software is designed to handle multiple tasks simultaneously, including sensor initialization, data acquisition, signal processing, display control, and wireless communication. The program continuously reads raw data from the pulse oximeter sensor and applies necessary algorithms to calculate important health parameters such as heart rate and blood oxygen saturation (SpO_2).

A key aspect of the software is its ability to establish and maintain communication between the microcontroller and external platforms using wireless protocols. The integration of Wi-Fi functionality allows the system to send real-time data to cloud-based services, where it can be stored, analyzed, and visualized in graphical form. This enables users and healthcare professionals to access patient data from any location, improving convenience and response time in case of emergencies.

Another important feature of the software is its capability to implement condition-based logic. The program is designed to compare the measured values with predefined threshold limits. If any parameter exceeds or falls below the safe range, the system triggers alerts, thereby ensuring early detection of potential health issues. This enhances the reliability and effectiveness of the monitoring system, making it suitable for continuous healthcare applications.

The software environment also includes development tools and libraries that simplify coding and improve system efficiency. These tools provide pre-defined functions for sensor interfacing, display management, and network communication, reducing development complexity. Additionally, debugging tools such as serial monitoring help in testing the system, identifying errors, and ensuring accurate performance before deployment.

Furthermore, the software is designed to be flexible and scalable, allowing future enhancements such as integration of additional sensors, mobile applications, and advanced data analytics. This adaptability ensures that the system can evolve with technological advancements and meet future healthcare demands. The program continuously reads raw data from the pulse oximeter sensor and applies necessary algorithms to calculate important health parameters such as heart rate and blood oxygen saturation (SpO_2).

In summary, the software requirements not only enable the functioning of the system but also add intelligence, connectivity, and reliability. The seamless interaction between software and hardware components results in a smart, efficient, and real-time health monitoring system capable of improving patient care and safety. These values are then formatted and displayed on a local output device while also being transmitted to a cloud platform for remote monitoring.

4.2.1 ThingSpeak Cloud Platform

ThingSpeak is an open-source Internet of Things (IoT) cloud platform that plays a crucial role in the proposed cardio-respiratory monitoring system. It is primarily used for collecting, storing, analyzing, and visualizing real-time data generated by the hardware components. The platform provides an efficient way to monitor health parameters remotely, making it highly suitable for modern healthcare applications. The pulse oximeter and EMG sensor enables real-time interaction with the microcontroller, which helps in monitoring outputs instantly during development and testing phases.

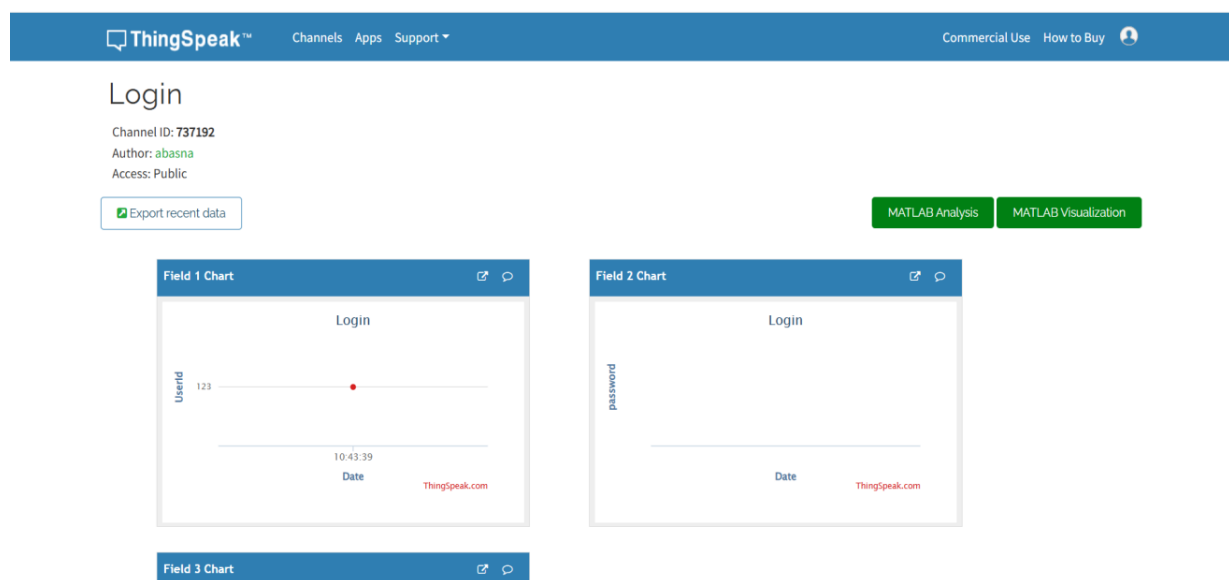


Figure 4.2.1 ThinkSpeak Platform

In this system Figure 4.2.1, the NodeMCU microcontroller transmits sensor data such as heart rate and blood oxygen saturation (SpO_2) to the ThingSpeak server using Wi-Fi connectivity. The data is sent through HTTP protocols using a unique API key assigned to each user channel. This ensures secure and organized data transfer from the device to the cloud.

Once the data is received, ThingSpeak stores it in dedicated channels where each parameter is assigned a specific field. These fields allow structured data storage, making it easier to analyze and retrieve information. The platform automatically updates the data at regular intervals, enabling continuous monitoring of the patient's health condition. One of

the key features of ThingSpeak is its ability to provide graphical visualization of data. It generates real-time graphs such as Heart Rate vs Time and SpO₂ vs Time, which help in understanding trends and variations over a period. These visual representations are highly useful for doctors and users to quickly interpret the patient's condition without analyzing raw data. ThingSpeak also supports data analytics and processing through built-in MATLAB integration. This allows advanced analysis such as filtering, prediction, and anomaly detection. By applying these analytical, system can be further enhanced early warnings for critical health conditions.

The platform is accessible from anywhere using internet-enabled devices such as smartphones, tablets, or computers. This remote accessibility makes it convenient for healthcare professionals to monitor multiple patients simultaneously without being physically present. It also benefits patients by enabling home-based health monitoring. In addition, ThingSpeak provides data logging and historical data storage, which can be used for long-term health analysis. Doctors can review past data trends to diagnose conditions more accurately and suggest appropriate treatments. This feature enhances the overall effectiveness of the healthcare monitoring system.

Furthermore, ThingSpeak is easy to use and integrate with microcontrollers like NodeMCU. It provides a simple interface for creating channels, generating API keys, and customizing dashboards. Its compatibility with IoT devices and support for real-time data handling make it a reliable choice for this project. In conclusion, the ThingSpeak cloud platform acts as a powerful backbone for remote monitoring, data visualization, and analysis in the proposed system. Its features such as real-time data updates, graphical representation, alert generation, and remote accessibility significantly enhance the functionality and usability of the IoT-based health monitoring system.

4.2.2 Firebase Database

Firebase Realtime Database is used as the central cloud storage and synchronization platform in the proposed IoT-based wearable safety system, as shown in Fig. 4.2.2. It plays a crucial role in enabling real-time communication between the hardware unit (ESP32) and the software interfaces such as the web dashboard and mobile monitoring system. These

database continuously stores, updates, and retrieves physiological sensor data collected from wearable device, ensuring that health information is always available for real-time monitoring and analysis. The system uploads key physiological parameters such as heart rate, blood oxygen saturation (SpO₂), EMG muscle activity, and calculated stress/fear levels to the Firebase cloud in real time. These values are transmitted from the ESP32 microcontroller using Wi-Fi connectivity through API or HTTP-based requests. Once the data reaches Firebase, it is instantly stored and synchronized across all connected devices.

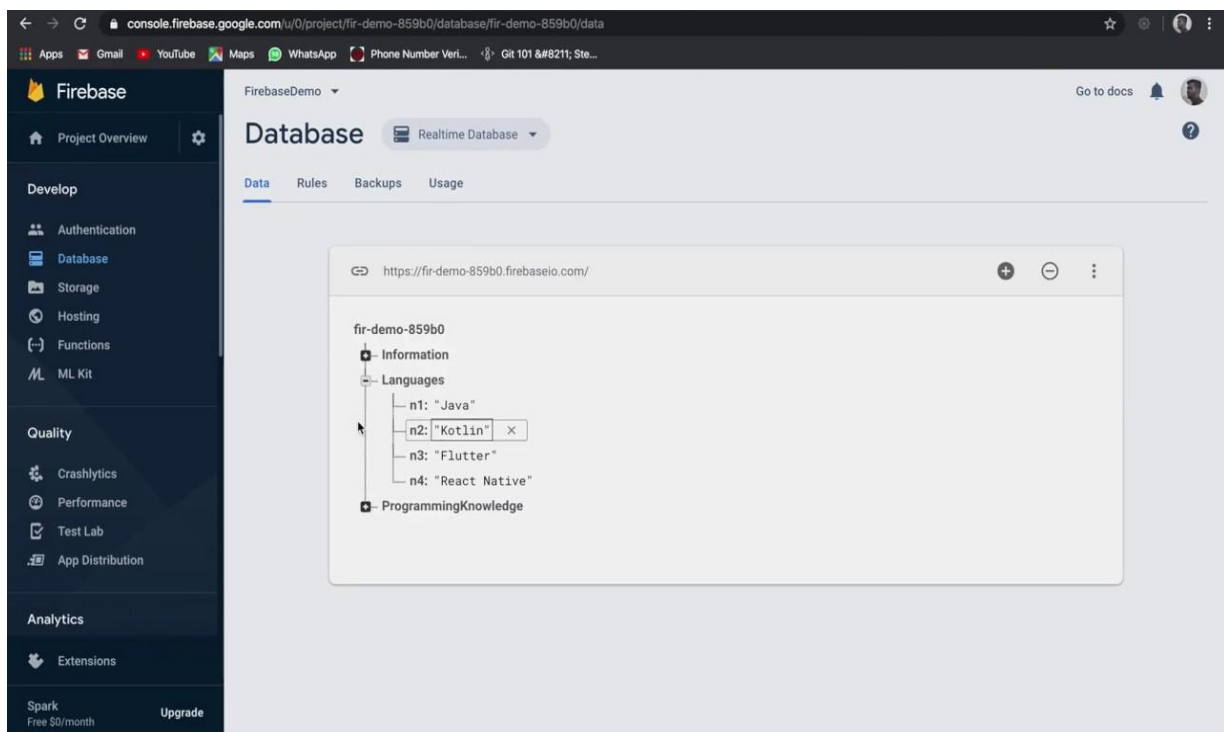


Figure 4.2.2 Firebase Realtime Database

Firebase Realtime Database is a cloud-hosted NoSQL database provided by Google. It allows data to be stored and synchronized in real time across multiple devices. In this project, Firebase can be used to store sensor data and instantly update it on connected applications such as mobile apps or web dashboards. The database stores data in JSON format, making it flexible and easy to manage. One of the key advantages of Firebase is its real-time synchronization feature. Whenever new data is uploaded, it is immediately reflected on all connected devices. It also provides strong security features, including authentication and access control.

Furthermore, Firebase also supports historical data logging, which allows long-term storage and analysis of physiological trends. This feature can be useful for future enhancements such as health prediction, anomaly detection, and machine learning-based monitoring systems. Firebase is highly scalable and suitable for advanced applications where real-time updates. Overall, Firebase Realtime Database significantly improves the efficiency, fast, secure, and real-time cloud communication.

4.2.3 User Authentication Interface

The User Authentication Interface is an important component of the IoT-based cardio-respiratory monitoring system, as it ensures secure access to the system and protects sensitive health data. Since the system involves storing and transmitting personal medical information such as heart rate and SpO₂ levels, it is essential to restrict access only to authorized users such as patients, doctors, or caregivers.

The authentication interface in Figure 4.2.3 provides a secure login mechanism where users must enter valid credentials, such as a username and password, to access the system. These credentials are verified against a stored database in the cloud platform. Only if the entered details match the stored records, the user is granted access to the monitoring dashboard. This process helps in preventing unauthorized access and ensures data privacy. The interface is typically designed as a web-based or mobile-based login page with a simple and user-friendly layout. It includes input fields for username/email and password, along with options such as “Login,” “Register,” and “Forgot Password.” The registration feature allows new users to create an account by providing necessary details, which are then securely stored in the database.

One of the key features of the user authentication interface is secure data handling. Passwords are usually encrypted before being stored in the database to prevent misuse. Authentication tokens or session IDs are generated after successful login, which helps maintain user sessions securely without repeatedly asking for credentials. The interface can also support role-based access control. For example, a patient may only view their own health data, while a doctor can monitor multiple patients. This ensures proper data management and controlled access based on user roles.

In addition, the authentication system may include multi-factor authentication (MFA), where users are required to verify their identity using an additional method such as a one-time password (OTP) sent to their mobile device or email. This adds an extra layer of security to the system. The integration of the authentication interface with the cloud database ensures that user data and login credentials are synchronized and updated in real time. This allows users to access their accounts from different devices while maintaining security. Furthermore, the authentication interface improves the overall usability of the system by providing personalized access. Users can log in to view their specific health data, track history, and receive alerts. This makes the system more interactive and user-centric.

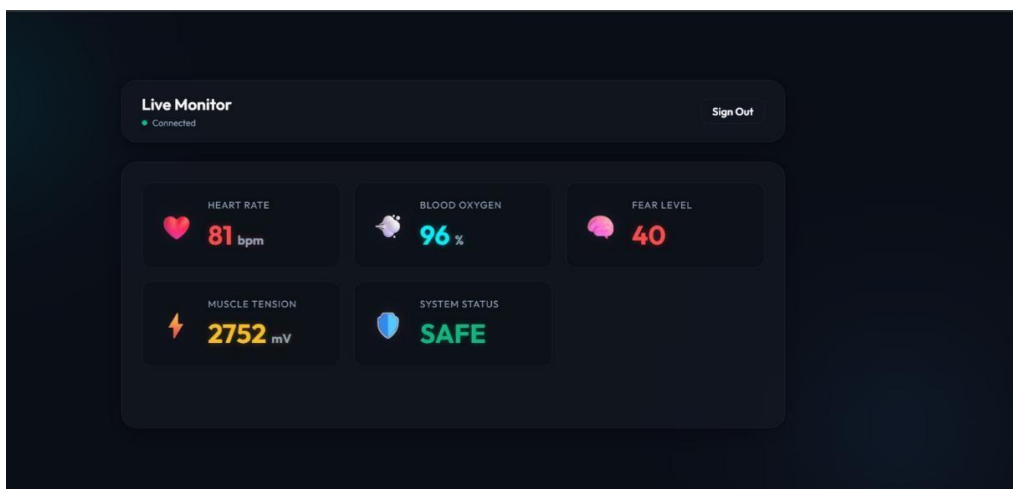


Figure 4.2.3 Live Monitoring Dashboard

One of the main features of the live monitoring dashboard is real-time data visualization. It presents continuous updates of health parameters through line graphs, charts, and gauges. For example, graphs such as Heart Rate vs Time and SpO₂ vs Time allow users to observe trends and variations over a period. This helps in identifying abnormal patterns or sudden changes in the patient's condition. The dashboard also includes numerical displays that show the current values of heart rate and SpO₂. These values are updated instantly as new data is received, providing immediate feedback to the user. In addition, color indicators or warning messages can be used to highlight abnormal readings displayed in red, indicating a potential health risk.

Another important feature is the alert and notification system integrated within the dashboard. When the monitored parameters exceed predefined threshold levels, the dashboard generates alerts to notify the user. These alerts can be in the form of visual warnings, sound notifications, or messages sent to connected devices. This ensures timely action in case of emergencies. The dashboard is designed to be accessible from multiple devices such as smartphones, tablets, and computers.

This remote accessibility allows healthcare professionals to monitor patients from any location, improving efficiency and response time. It also enables patients to track their own health conveniently at home. User-friendly design is an essential aspect of the dashboard. It is typically developed with a simple layout, clear labels, and easy navigation. This ensures that even non-technical users can understand and operate the system without difficulty. Features such as login authentication and personalized views enhance usability and security. Additionally, the dashboard may include historical data analysis, where users can view past records and trends. This helps in long-term health assessment and supports better medical decision-making. The integration with cloud databases ensures that all data is stored and updated continuously.

The dashboard also includes numerical displays that show the current values of heart rate and SpO₂. These values are updated instantly as new data is received, providing immediate feedback to the user. These alerts can be in the form of visual warnings, sound notifications, or messages sent to connected devices. This ensures timely action in case of emergencies.

The dashboard can be developed using web technologies such as HTML, CSS, and JavaScript, or through IoT platforms like ThingSpeak, which provide built-in visualization tools. Advanced implementations may include mobile applications with interactive interfaces and real-time notifications. In conclusion, the Live Monitoring Dashboard plays a crucial role in transforming raw sensor data into meaningful and actionable information. It enhances the effectiveness of the IoT-based monitoring system by providing real-time visualization, remote accessibility, and alert mechanisms, thereby improving patient care and safety.

CHAPTER 5

RESULTS AND DISCUSSION

This chapter presents the results obtained from the implementation of the IoT-based cardio-respiratory monitoring system and provides a detailed discussion on its performance. The system was designed to measure vital health parameters such as heart rate and blood oxygen saturation (SpO₂) in real time and transmit the data to a cloud platform for remote monitoring. The results demonstrate the effectiveness, accuracy, and reliability of the system under different conditions Figure. 5.1.

The experimental results demonstrate that the system is capable of delivering accurate real-time physiological data with minimal delay, making it suitable for wearable health monitoring and emergency detection applications.

5.1 HARDWARE PROTOTYPE



Figure 5.1 Hardware Prototype of IoT-Based Wearable Safety System

Figure. 5.1 The hardware prototype of the proposed IoT-based cardio-respiratory monitoring system was successfully designed and implemented using essential electronic components such as NodeMCU (ESP8266), MAX30100/MAX30102 pulse oximeter

sensor, OLED display, and power supply. The prototype was assembled on a breadboard using connecting wires to ensure flexibility during testing and modifications. The pulse oximeter sensor was interfaced with the NodeMCU using I2C communication protocol, enabling efficient data transfer. The OLED display was also connected via I2C, allowing real-time visualization of measured parameters such as heart rate and SpO₂ levels. The entire system was powered using a 5V USB supply, ensuring stable operation.

During the testing phase, the hardware prototype demonstrated proper functioning, where the sensor accurately detected physiological signals when a user placed their fingertip on it. The NodeMCU processed the sensor data and displayed the results instantly on the OLED screen. The compact design of the prototype makes it suitable for portable and wearable healthcare applications. The prototype also showed good integration between hardware components, with minimal signal loss and stable communication. The use of a breadboard allowed easy troubleshooting and quick modifications during development. Overall, the hardware prototype validated the feasibility of the proposed system and confirmed its ability to perform real-time health monitoring.

The pulse oximeter sensor, which is the primary sensing unit, was interfaced with the NodeMCU using the I2C communication protocol. This ensured efficient and reliable data transmission between the sensor and the microcontroller. Similarly, the OLED display was also connected via I2C, enabling the display of real-time physiological parameters such as heart rate and blood oxygen saturation (SpO₂). The use of I2C communication reduced wiring complexity and improved overall system performance.

The NodeMCU acted as the central processing unit, responsible for reading the sensor data, processing it using embedded algorithms, and converting it into meaningful values. These values were then displayed on the OLED screen and transmitted to the cloud platform via Wi-Fi. The system was powered using a 5V USB supply, ensuring stable voltage and continuous operation throughout the testing phase. During the experimental setup, the user placed their fingertip on the pulse oximeter sensor. The sensor detected variations in blood flow and oxygen levels using light absorption techniques.

5.2 LIVE MONITORING SYSTEM

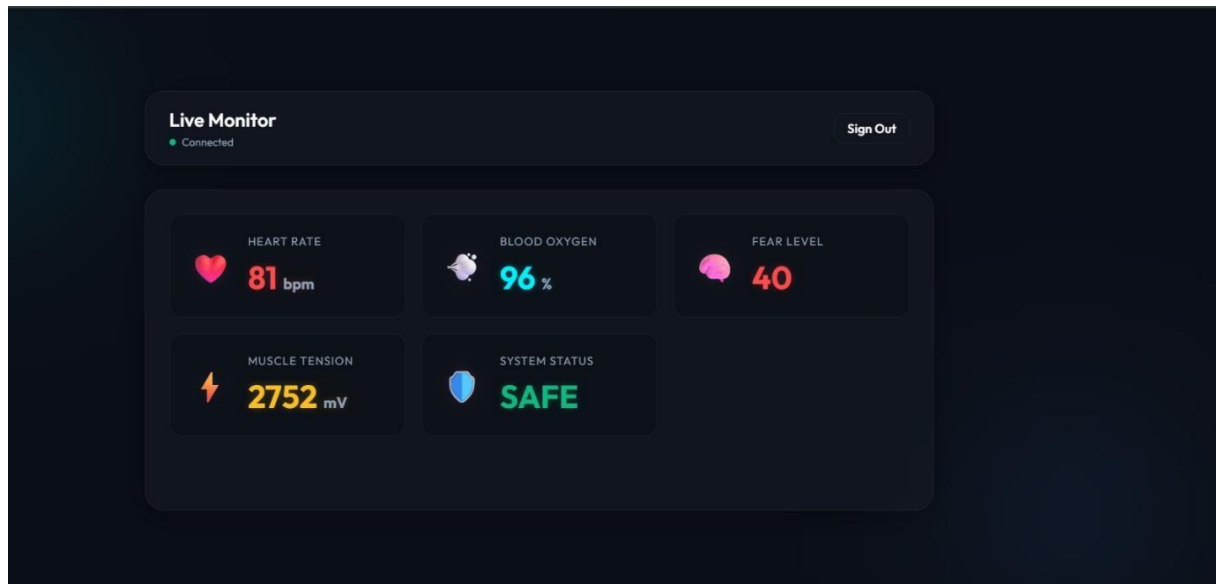


Figure 5.2 Live Monitoring Dashboard Displaying Real-Time Physiological Parameters

The Figure 5.2 live monitoring system is a key feature of the proposed project, enabling real-time observation and analysis of physiological parameters through cloud integration. The system utilizes the Wi-Fi capability of the NodeMCU to transmit sensor data to the cloud platform, where it is stored, processed, and visualized.

Once the sensor data is acquired and processed by the NodeMCU, it is transmitted to the cloud platform (ThingSpeak) using HTTP protocols and API keys. The cloud platform receives this data and stores it in structured fields within a dedicated channel. Each parameter, such as heart rate and SpO_2 , is assigned a specific field, ensuring organized data storage.

The live monitoring dashboard displays the data in both numerical and graphical formats. The numerical display shows the current values of heart rate and SpO_2 , while the graphical representation provides a visual understanding of how these values change over time. Graphs such as Heart Rate vs Time and SpO_2 vs Time are continuously updated, allowing users to track variations and trends. One of the major advantages of the live monitoring system is remote accessibility.

This feature is particularly useful for doctors and caregivers, as it allows them to monitor patients without being physically present. The system also includes a real-time alert mechanism. Threshold values are predefined for heart rate and SpO₂ levels. If the measured values exceed or fall below these limits, the system generates alerts. These alerts may be displayed on the dashboard or sent as notifications, enabling quick response to critical conditions.

During testing, the live monitoring system demonstrated reliable performance with continuous data updates and minimal delay. The Wi-Fi communication was stable, and the data was successfully transmitted to the cloud without interruption. The graphical visualization provided clear insights into the patient's health condition, making it easier to interpret the data. The integration of cloud technology enhances the functionality of the system by enabling data storage, remote monitoring, and long-term analysis. It also provides scalability, allowing the system to handle multiple users and large amounts of data. Overall, the live monitoring system significantly improves the efficiency and usability of the healthcare monitoring system.

5.3 DISCUSSION

The results obtained from the implementation of the IoT-based cardio-respiratory monitoring system demonstrate that the system is capable of providing accurate, reliable, and real-time health monitoring. The integration of hardware components with software and cloud platforms has resulted in a well-functioning system that meets the objectives of the project. The hardware prototype performed efficiently, with the pulse oximeter sensor accurately detecting heart rate and SpO₂ levels.

The NodeMCU successfully processed the data and ensured smooth communication between the sensor, display, and cloud platform. The OLED display provided immediate feedback, allowing users to view their health parameters locally. The live monitoring system further enhanced the functionality by enabling remote access and real-time data visualization. The graphical representation of data made it easier to analyze trends and identify abnormalities. The system was able to detect both normal and abnormal conditions, proving in continuous health monitoring generation.

This feature is particularly important in healthcare applications, as it enables timely medical intervention and reduces the risk of serious complications. The use of cloud technology ensures that data is stored securely and can be accessed anytime for further analysis. However, certain limitations were observed during the implementation. The accuracy of the sensor readings may be affected by factors such as improper finger placement, motion artifacts, and external light interference. Additionally, the system relies on stable internet connectivity for cloud communication, which may not always be available in remote areas.

Despite these limitations, the overall performance of the system is satisfactory. The advantages of real-time monitoring, remote accessibility, and low-cost implementation outweigh the limitations. The system provides a practical solution for continuous health monitoring in both clinical and home environments. Future improvements can be made by integrating additional sensors such as ECG, temperature, and blood pressure sensors to provide a more comprehensive health monitoring system.

The graphical visualization provided clear insights into the patient's health condition, making it easier to interpret the data. The integration of cloud technology enhances the functionality of the system by enabling data storage, remote monitoring, and long-term analysis. It also provides scalability, allowing the system to handle multiple users and large amounts of data.

Advanced data analysis techniques, including machine learning algorithms, can also be implemented to predict potential health risks. In conclusion, the project successfully demonstrates the application of IoT technology in healthcare. The integration of hardware components with software and cloud platforms has resulted in a well-functioning system that meets the objectives of the project. The results confirm that the system is efficient, reliable, and capable of improving patient care through continuous monitoring and early detection of health abnormalities.

CHAPTER 6

CONCLUSION

The project titled “Intelligent IoT-Based Multi-Parameter Cardio-Respiratory and Health Monitoring System with Smart Health Scoring” successfully demonstrates the design and development of a modern healthcare solution that integrates Internet of Things (IoT) technology with biosensors and cloud computing. The primary objective of this project was to create a system capable of continuously monitoring multiple physiological parameters such as heart rate, oxygen saturation (SpO₂), respiratory rate, blood pressure estimation, and stress levels in real time. This objective has been effectively achieved through the implementation of an efficient and reliable system that collects, processes, and transmits health data to a cloud platform for remote monitoring.

The proposed system addresses the major limitations of traditional healthcare monitoring methods, which rely heavily on periodic hospital visits and lack continuous observation capabilities. By enabling real-time monitoring, the system ensures that any abnormalities in the patient’s health condition can be detected at an early stage. This significantly reduces the risk of severe complications and supports timely medical intervention. The integration of IoT technology plays a crucial role in enhancing accessibility, as it allows healthcare professionals and caregivers to monitor patient data remotely through a cloud-based dashboard.

One of the key highlights of this project is the implementation of a smart health scoring mechanism. This feature simplifies complex medical data into a single, easy-to-understand score that represents the overall health condition of the patient. By combining multiple parameters into a unified metric, the system makes it easier for both patients and healthcare providers to quickly assess health status and take necessary actions. This approach not only improves usability but also enhances decision-making in critical situations. This significantly reduces the risk of severe complications and supports timely medical intervention.

The system also incorporates an alert mechanism that notifies users whenever any parameter exceeds predefined threshold values. This ensures immediate attention during emergency situations and improves patient safety. The ability to generate real-time alerts is particularly beneficial for patients with chronic diseases, elderly individuals, and those who require continuous monitoring. It reduces dependency on constant physical supervision and provides a sense of security to both patients and their families.

Another important achievement of this project is the development of a cost-effective and portable system. By using low-cost components such as sensors and the ESP8266 microcontroller, the system becomes affordable and accessible to a wide range of users. Its compact design and ease of use make it suitable for both hospital and home environments. This is especially advantageous for patients who face difficulty in visiting healthcare facilities frequently due to health conditions or geographical limitations.

The integration of cloud computing further enhances the functionality of the system by providing secure data storage and easy access to historical health records. This allows healthcare professionals to analyze trends and patterns in patient data, leading to better diagnosis and treatment planning. The ability to store and retrieve data efficiently also supports long-term health monitoring and preventive healthcare practices.

Despite its advantages, the system has certain limitations, such as dependency on internet connectivity, sensor accuracy constraints, and basic data analysis capabilities. However, these limitations can be addressed in future enhancements by incorporating advanced technologies such as artificial intelligence, machine learning, and improved sensor modules. The system has strong potential for further development and can be upgraded to include additional features such as predictive analysis, advanced diagnostics, and integration with smart healthcare ecosystems.

In conclusion, the proposed Intelligent IoT-Based Multi-Parameter Cardio-Respiratory and Health Monitoring System provides an effective, reliable, and scalable solution for modern healthcare challenges. It successfully combines real-time monitoring, remote accessibility, intelligent analysis, and user-friendly design to improve patient care

and safety. The system contributes to the advancement of smart healthcare technologies by reducing the need for frequent hospital visits, enabling early detection of health issues, and supporting continuous health monitoring. By enabling real-time monitoring, the system ensures that any abnormalities in the patient's health condition can be detected at an early stage. This significantly reduces the risk of severe complications and supports timely medical intervention. The ability to generate real-time alerts is particularly beneficial for patients with chronic diseases, elderly individuals, and those who require continuous monitoring.

6.1 FUTURE ENHANCEMENTS

The proposed “Intelligent IoT-Based Multi-Parameter Cardio-Respiratory and Health Monitoring System with Smart Health Scoring” can be further improved by incorporating several advanced features and technologies to enhance its performance, accuracy, and usability. One of the major future enhancements is the integration of Artificial Intelligence (AI) and Machine Learning (ML) algorithms. By using AI techniques, the system can analyze large volumes of health data and predict potential health risks at an early stage. Predictive analytics can help in identifying patterns and trends, enabling proactive healthcare and reducing the chances of critical conditions.

6.1.1 Integration of Artificial Intelligence and Machine Learning

One of the most significant future enhancements of the proposed system is the integration of Artificial Intelligence (AI) and Machine Learning (ML) techniques. By incorporating AI algorithms, the system can analyze large volumes of real-time and historical health data to identify patterns and predict potential health risks. Machine learning models can be trained to detect abnormalities and provide early warnings for diseases such as heart conditions, respiratory disorders, and hypertension. By enabling real-time monitoring, the system ensures that any abnormalities in the patient's health condition can be detected at an early stage. This predictive capability will transform the system from a monitoring tool into a smart decision-support system, enabling preventive healthcare and reducing the chances of critical emergencies.

6.1.2 Addition of Advanced Medical Sensors

The integration of a GPS module is an important future enhancement for improving the safety functionality of the wearable device. In emergency situations, knowing the exact location of the user is critical for quick response and rescue operations. By adding GPS tracking, the system can automatically send real-time location coordinates along with health alerts when abnormal conditions are detected. This feature can be connected with Firebase or Telegram bot notifications, enabling caregivers or emergency contacts to immediately locate the user. This enhancement is especially useful for elderly care, outdoor workers, and individuals working in high-risk environments, significantly improving overall safety and response efficiency.

6.1.3 Development of a Dedicated Mobile Application

A dedicated mobile application can be developed to improve accessibility and user interaction. The mobile app can provide real-time monitoring, graphical representation of health data, and easy access to historical records. It can also include features such as personalized health recommendations, reminders for medication, and instant alerts in case of abnormal conditions. A user-friendly interface will ensure that patients, especially elderly individuals, can easily understand and use the system without technical difficulties.

6.1.4 Enhanced Data Security and Privacy Mechanisms

Since the system deals with sensitive patient data, enhancing data security and privacy is a critical future improvement. Advanced security measures such as data encryption, secure authentication, and blockchain-based data storage can be implemented to prevent unauthorized access. By adding GPS tracking, the system can automatically send real-time location coordinates along with health alerts when abnormal conditions are detected. Machine learning models can be trained to detect abnormalities and provide early warnings for diseases such as heart conditions, respiratory disorders, and hypertension. These techniques will ensure the confidentiality and integrity of patient information, increasing user trust and making the system suitable for real-world healthcare deployment.

6.1.5 Scalability and Multi-Patient Monitoring

The system can be upgraded to support multiple patients simultaneously, making it suitable for hospitals, clinics, and healthcare organizations. By improving cloud scalability, the system can handle large volumes of data from multiple users without performance issues. This will enable centralized monitoring of patients and efficient management of healthcare resources. Healthcare providers can track multiple patients in real time using a single platform.

6.1.6 Energy Efficiency and Power Optimization

Future enhancements can focus on improving the energy efficiency of the system. By using low-power sensors and optimizing the microcontroller's power consumption, the system can achieve longer battery life. This is particularly important for wearable and portable devices that require continuous operation. Incorporating rechargeable batteries and energy-saving techniques will enhance the reliability and usability of the system.

6.1.7 Integration with GPS and Emergency Response Systems

The system can be enhanced by integrating GPS tracking and emergency response features. In case of critical health conditions, the system can automatically send alerts along with the patient's real-time location to healthcare providers or emergency services. This will ensure quick medical assistance and improve response time during emergencies. This feature is especially beneficial for elderly patients and individuals living alone.

6.1.8 Cloud Computing and Big Data Analytics

Future versions of the system can utilize advanced cloud computing technologies and big data analytics to process large datasets efficiently. By analyzing long-term health data, the system can identify trends and provide valuable insights for better diagnosis and treatment planning. It can also include features such as personalized health recommendations, reminders for medication, and instant alerts in case of abnormal conditions. Big data analytics can also support medical research by providing real-time health data for analysis and study.

6.1.9 Voice Assistant and Smart Interaction

The system can be integrated with voice assistant technologies to provide voice-based interaction and feedback. Patients can receive health updates, alerts, and instructions through voice commands, making the system more user-friendly. This feature is particularly useful for elderly and visually impaired individuals who may find it difficult to use traditional interfaces.

6.1.10 Integration with Smart Healthcare Ecosystem

In the future, the system can be integrated with other smart healthcare devices and hospital management systems to create a complete healthcare ecosystem. Data from various devices can be combined and analyzed to provide a holistic view of the patient's health. This integration will improve coordination between different healthcare services and enhance overall patient care.

6.1.11 Wearable and Compact Device Development

The system can be further miniaturized into a wearable device such as a wristband or patch. This will improve comfort and usability for continuous monitoring. It can also include features such as personalized health recommendations, reminders for medication, and instant alerts in case of abnormal conditions. Wearable devices will enable patients to carry the system monitor their health without any inconvenience.

6.1.12 Real-Time Video Consultation Integration

Future enhancements can include integration with telemedicine platforms that support real-time video consultation. Patients can directly communicate with doctors through the system, sharing their live health data during consultations. This will improve the effectiveness of remote healthcare services. This feature can be connected with Firebase or Telegram bot notifications, enabling caregivers or emergency contacts to immediately locate the user. Patients can receive health updates, alerts, and instructions through voice commands, making the system more user-friendly. This is particularly important for wearable and portable devices that require continuous operation.

In conclusion, the proposed system has significant potential for further development and improvement. By incorporating advanced technologies such as AI, additional sensors, enhanced security, and smart integrations, the system can evolve into a highly intelligent and comprehensive healthcare solution. These future enhancements will not only improve the system's performance and reliability but also expand its applications in modern healthcare, making it a vital tool for smart and connected medical environments.

6.2 LIMITATIONS OF THE PROPOSED WORK

Although the proposed “Intelligent IoT-Based Multi-Parameter Cardio-Respiratory and Health Monitoring System with Smart Health Scoring” provides an efficient and modern solution for healthcare monitoring, it still has certain limitations that need to be addressed for improved performance and real-world implementation.

6.2.1 Key Limitations:

- Limited Sensor Accuracy
- Dependency on Internet Connectivity
- Power Supply Constraints
- Data Security and Privacy Issues
- Limited Clinical Validation
- Restricted Data Analysis Capability
- Hardware Complexity and Maintenance
- Scalability Challenges

6.2.2 Explanation of Limitations

One of the major limitations of the system is the limited accuracy of sensors. The system uses cost-effective sensors, which may not provide the same level of precision as clinical-grade medical devices. Factors such as improper sensor placement, patient movement, and environmental conditions can lead to variations in readings.

Another important limitation is the dependency on internet connectivity. The system relies on Wi-Fi to transmit data from the ESP8266 microcontroller to the cloud platform. In case of poor or unstable network conditions, data transmission may be delayed or interrupted, affecting real-time monitoring and alert generation. This can be a significant challenge in rural or remote areas with limited internet access.

The system also depends on a continuous power supply for proper functioning. Any power interruption can cause the system to stop working, leading to loss of real-time monitoring. Although battery backup can be used, it increases system complexity and requires regular maintenance. Data security and privacy is another concern, as sensitive patient information is transmitted and stored on cloud platforms. Without proper security measures such as encryption and secure authentication, there is a risk of unauthorized access or data breaches. Ensuring strong data protection mechanisms is essential for safe usage.

The system currently has limited clinical validation, meaning it may not yet meet all medical standards required for professional healthcare use. Extensive testing and certification are needed before it can be widely adopted in hospitals and clinical environments. Another limitation is the restricted data analysis capability. While the system provides a smart health score, it lacks advanced analytical features such as predictive analysis and AI-based diagnosis. This limits its ability to provide deeper insights into patient health conditions. One of the major limitations of the proposed system is the dependency on stable internet connectivity for real-time cloud monitoring and data transmission. In areas with poor Wi-Fi or network coverage, the system may experience delays in updating health parameters, which can affect continuous monitoring and emergency response. Since the device relies on IoT communication, any interruption in connectivity may reduce the reliability of remote healthcare services.

Another limitation is the accuracy of sensor readings under different environmental and user conditions. Sensors such as pulse oximeters, heart rate sensors, and temperature sensors may produce inaccurate values due to improper placement, body movement, sweating, or external noise. These variations can affect the overall Smart Health Score and may require calibration or advanced filtering techniques for precise measurement.

The proposed system is mainly designed for basic health monitoring and does not replace professional medical equipment or clinical diagnosis. Parameters such as blood pressure estimation and stress detection are calculated using indirect methods, which may not provide highly accurate medical-grade results. Therefore, the system should be considered as a supportive health monitoring tool rather than a complete diagnostic solution.

The hardware complexity and maintenance can also be a challenge, especially for non-technical users. Proper setup, calibration, and troubleshooting may require technical knowledge, making it less user-friendly for some patients. Finally, the system may face scalability challenges when used for monitoring multiple patients simultaneously. By incorporating advanced technologies such as AI, additional sensors, enhanced security, and smart integrations, the system can evolve into a highly intelligent and comprehensive healthcare solution. Factors such as improper sensor placement, patient movement, and environmental conditions can lead to variations in readings. Handling large volumes of data and maintaining system performance may require more advanced cloud infrastructure and optimization techniques.

In conclusion, while the proposed system offers significant advantages in real-time health monitoring and remote accessibility, addressing these limitations through improved sensor technology, better connectivity solutions, enhanced security, and advanced data analysis techniques can further enhance its effectiveness and reliability in future implementations.

6.3 SUMMARY OF THE PROJECT

The project titled “Intelligent IoT-Based Multi-Parameter Cardio-Respiratory and Health Monitoring System with Smart Health Scoring” focuses on developing a smart, efficient, and cost-effective solution for continuous health monitoring using modern Internet of Things (IoT) technology. The main objective of this system is to monitor multiple vital physiological parameters such as heart rate, oxygen saturation (SpO2), blood pressure estimation, respiratory rate, and stress level in real time, and to provide a comprehensive health assessment through a smart health score. This system is designed to address the

limitations of traditional healthcare monitoring methods, which require frequent hospital visits and do not support continuous observation of patient health.

In the proposed system, various biosensors are used to collect real-time data from the human body. These sensors are connected to an ESP8266 microcontroller, which acts as the central processing unit. The microcontroller collects the sensor data, processes it, and transmits it to a cloud platform using its built-in Wi-Fi capability. The cloud platform serves as a centralized storage system where patient data is securely stored and can be accessed remotely by healthcare professionals, caregivers, or family members through a web-based dashboard. This enables continuous monitoring of patient health from any location, improving accessibility and convenience.

A significant feature of the system is the implementation of a smart health scoring mechanism, which evaluates multiple health parameters and generates an overall health score. This score simplifies complex medical data into an easy-to-understand format, allowing quick assessment of the patient's condition. In addition, the system includes an alert mechanism that notifies users when any parameter exceeds predefined threshold limits.

The system is designed to be portable, user-friendly, and affordable, making it suitable for both hospital and home environments. It is especially beneficial for elderly individuals, patients with chronic diseases, and those who require continuous monitoring but cannot frequently visit healthcare facilities. By enabling real-time monitoring, remote access, and early detection of abnormalities, the system reduces the risk of severe health complications and improves overall healthcare efficiency. The main objective of this system is to monitor multiple vital physiological parameters such as heart rate, oxygen saturation (SpO₂), blood pressure estimation, respiratory rate, and stress level in real time, and to provide a comprehensive health assessment through a smart health score.

The main purpose of this project is to provide continuous health monitoring and instant access to patient data from anywhere. The measured values are displayed on a mobile application or web dashboard, enabling users and healthcare providers to track health conditions remotely. The system can also generate alerts when abnormal readings are

detected, helping in early diagnosis and emergency response. This project improves healthcare accessibility, reduces manual monitoring, and supports real-time medical observation. It is especially useful for elderly people, patients with chronic diseases, and remote healthcare applications. The Smart Health Monitoring System is cost-effective, portable, and easy to use, making it suitable for modern digital healthcare environments.

Overall, this project demonstrates the effective integration of IoT, biosensors, and cloud computing to create an intelligent healthcare monitoring system. It provides a reliable and scalable solution for modern healthcare challenges, contributing to improved patient care, reduced healthcare costs, and the advancement of smart healthcare technologies. The system uses sensors to measure vital signs such as heart rate, body temperature, oxygen level (SpO₂), and respiratory rate.

The system can also generate alerts when abnormal readings are detected, helping in early diagnosis and emergency response. The main objective of this system is to monitor multiple vital physiological parameters such as heart rate, oxygen saturation (SpO₂), blood pressure estimation, respiratory rate, and stress level in real time, and to provide a comprehensive health assessment through a smart health score.

In this system, various biomedical sensors are connected to a microcontroller unit such as Arduino Uno or ESP8266. These sensors collect health-related data from the human body and send the information to the controller for processing. The processed data is then transmitted to a cloud server or IoT platform through Wi-Fi communication. Doctors, caregivers, and users can access the health data remotely through a mobile application or web dashboard. This feature helps in continuous patient monitoring without requiring physical presence in hospitals or clinics.

The project mainly focuses on improving healthcare efficiency by providing real-time monitoring and automatic alert generation during abnormal health conditions. If the measured values exceed the normal range, the system can send warning notifications or emergency alerts to family members or healthcare professionals. This helps in taking immediate medical action and reduces the risk of severe health complications.

Advanced technologies such as Artificial Intelligence (AI), machine learning, and cloud analytics can be added for predictive healthcare analysis and disease detection. The project demonstrates how IoT technology can transform traditional healthcare systems into smart digital healthcare environments by enabling continuous monitoring, remote access, and instant alert mechanisms. This system contributes to better patient care, improved medical response time, and enhanced healthcare management in modern society. Additional sensors and features can also be integrated in the future to enhance system performance. These sensors are connected to a microcontroller like Arduino or ESP8266, it to a cloud through Wi-Fi technology.

Furthermore, the project supports the concept of smart hospitals and digital healthcare ecosystems by integrating biosensors, wireless communication, cloud computing, and intelligent data processing into a single platform. The system demonstrates how modern technologies can work together to provide efficient, affordable, and user-friendly healthcare solutions. By reducing healthcare costs, improving patient monitoring accuracy, and enabling faster medical response, the proposed system contributes significantly to the advancement of modern healthcare technology.

The compact and portable design of the system also makes it suitable for wearable healthcare applications. Patients can use the device comfortably in their daily activities while continuously monitoring their health conditions. This portability increases user convenience and ensures uninterrupted monitoring, especially for elderly patients and individuals with chronic illnesses who require regular observation. The lightweight design and wireless communication capabilities make the system practical for both indoor and outdoor healthcare monitoring applications.

CHAPTER 7

REFERENCES

1. Abdulmalek S., Nasr A., Ahmed M. A., Khalifa K. S., “IoT-Based Healthcare Monitoring System: A Review,” IEEE Access, 2022.
2. Ali F., El-Sappagh S., Ali A., Kwak K., “Cloud-Based IoT Health Monitoring System,” Journal of Healthcare Engineering, 2023.
3. Amir A., Usman M., Raza H., Khan S., “IoT-Based Health Monitoring Using Arduino and FPGA,” 2025.
4. Baraeinejad B., Shokouhifar M., Jalali A., “Design and Implementation of an IoT-Based Respiratory Motion Sensor,” Engineering Proceedings, vol. 118, no. 1, 2025.
5. Chauhan R., Singh P., Verma A., Tiwari S., “AI-Powered IoT Framework for Real-Time Respiratory Monitoring,” Springer, 2026.
6. Chen L., Zhang X., Liu Y., Wang Z., “Remote Patient Monitoring Using IoT and AI,” 2023.
7. Dheman K., Sharma R., Gupta A., Mehta P., “i-CardiAx: Wearable IoT-Driven System for Vital Sign Monitoring,” arXiv, 2024.
8. Georgieva-Tsaneva G., Gancheva V., Ruskova I., Cheshmedzhiev D., “Healthcare Monitoring Using an Internet of Things-Based Cardio System,” IoT Journal, vol. 6, no. 1, 2025.
9. Georgieva-Tsaneva G., Gancheva V., Ruskova I., “HRV Analysis in IoT-Based Cardiac Monitoring Systems,” 2025.
10. Gupta A., Kumar R., Jain S., Sharma P., “IoT-Based Smart Patient Monitoring System,” 2022.

11. Harez J. B., Ahmed M. S., Hassan A. K., Ibrahim R. K., "IoT-Driven System for Continuous Monitoring of Heart Disease Patients Post-Surgery," arXiv, 2024.
12. Jose J. P., Varghese R., Thomas M., Philip S., "IoT Enabled Health Monitoring System Using Optimization and Deep Learning," Scientific Reports, vol. 15, 2025.
13. Khan M., Rehman S., Ullah A., Ali T., "IoT-Based Wearable Health Monitoring Devices," 2022.
14. Kumar S., Singh R., Verma P., Sharma A., "Smart IoT-Based Healthcare Monitoring System," IEEE Access, 2023.
15. Mamun M., Hossain S., Islam M., Roy N., "IoT Environmental and Health Monitoring Devices," 2024.
16. Mohapatra A. G., Nayak S., Das P., Mishra R., "IoT-Driven Remote Health Monitoring System," ScienceDirect, 2025.
17. Mohapatra A., Nayak S., Das P., "Sensor Fusion in IoT Healthcare Systems," 2025.
18. Nayab S., Arif M., Khan A., Iqbal F., "IoT Framework for Cardiovascular Monitoring," arXiv, 2025.
19. Patel V., Shah N., Mehta R., Desai S., "Smart Healthcare Monitoring Using IoT Sensors," 2022.
20. Paul S. K., Hasan M., Islam R., Rahman T., "IoT-Based Real-Time Healthcare Monitoring with Deep Learning," 2024.
21. Pradeesh D. A., Subiramaniam N. P., "IoT-Enabled Smart Health Monitoring System with Deep Learning Models," International Journal of Engineering Trends and Technology, vol. 73, no. 1, 2025.
22. Sharma P., Gupta R., Singh A., Verma M., "Real-Time ECG Monitoring Using IoT," 2022.

23. Singh R., Yadav S., Mishra A., Tiwari P., “IoT-Based Smart Medical Monitoring Using Machine Learning,” 2023.
24. Spachos P., Song L., Hatzinakos D., “IoT Wearable Systems for Health Monitoring,” 2024.
25. Zhang Y., Wang H., Liu X., Chen Z., “Wearable IoT Sensors for Cardio-Respiratory Monitoring,” IEEE Sensors Journal, 2023.